EYEL-S3000ABR 8ch SVM

Rev.	Date	Description	Page	Remark
V01	2026.03.07	- Initial		

Title
00.TITLE

Size	Document Number	Rev
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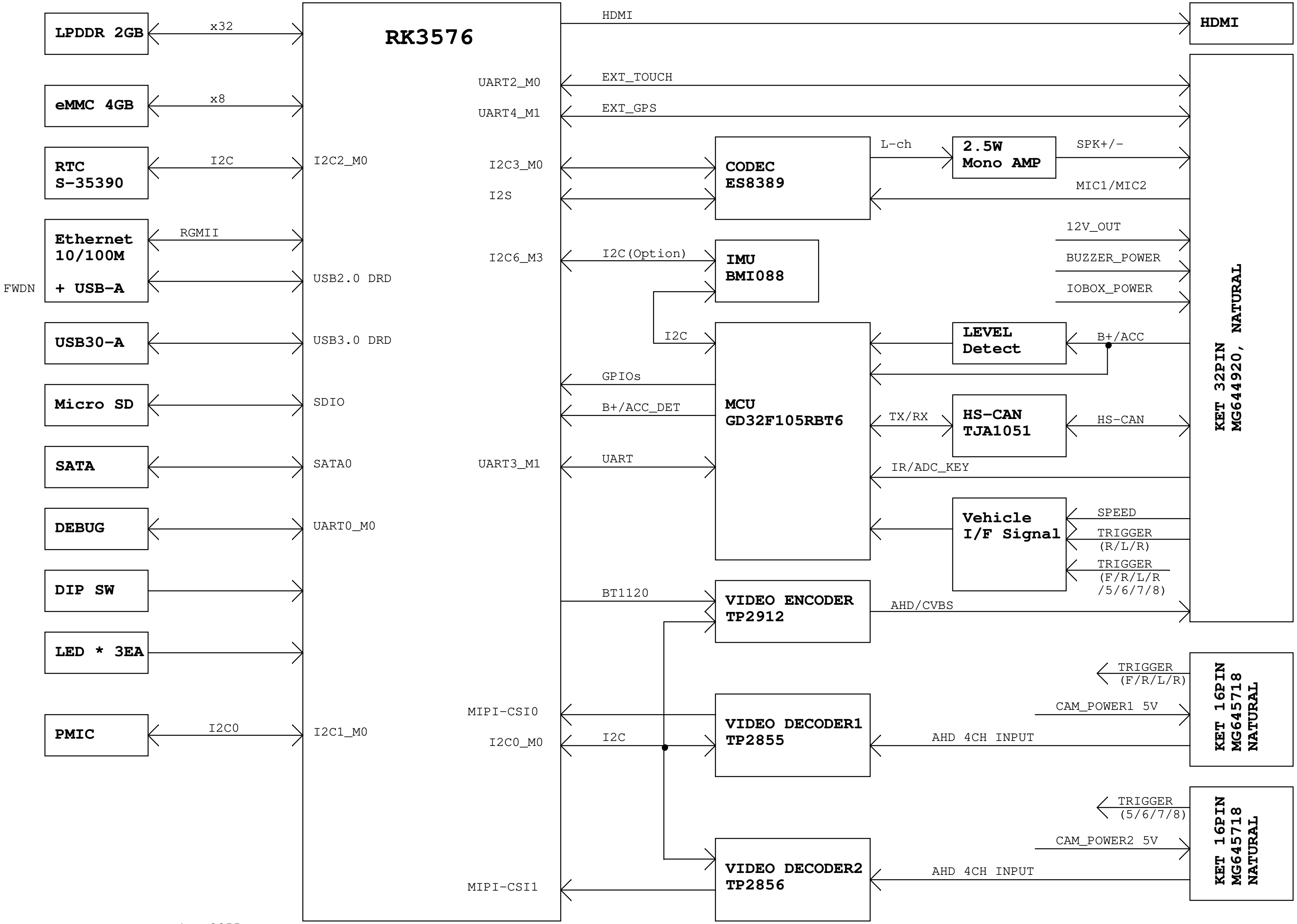

I2C0_M0	VIDEO DECODER1 TP2855 VIDEO DECODER2 TP2855 VIDEO ENCODER TP2912
I2C1_M0	PMIC RK806S-5
I2C2_M0	RTC S-35390A
I2C3_M0	CODEC ES8389
I2C6_M3 (Option)	BMI088
I2C_MCU	BMI088

UART0_M0	DEBUG
UART2_M0	External TOUCH
UART3_M1	MCU
UART4_M1	External GPS

Title **01.Block Diagram**

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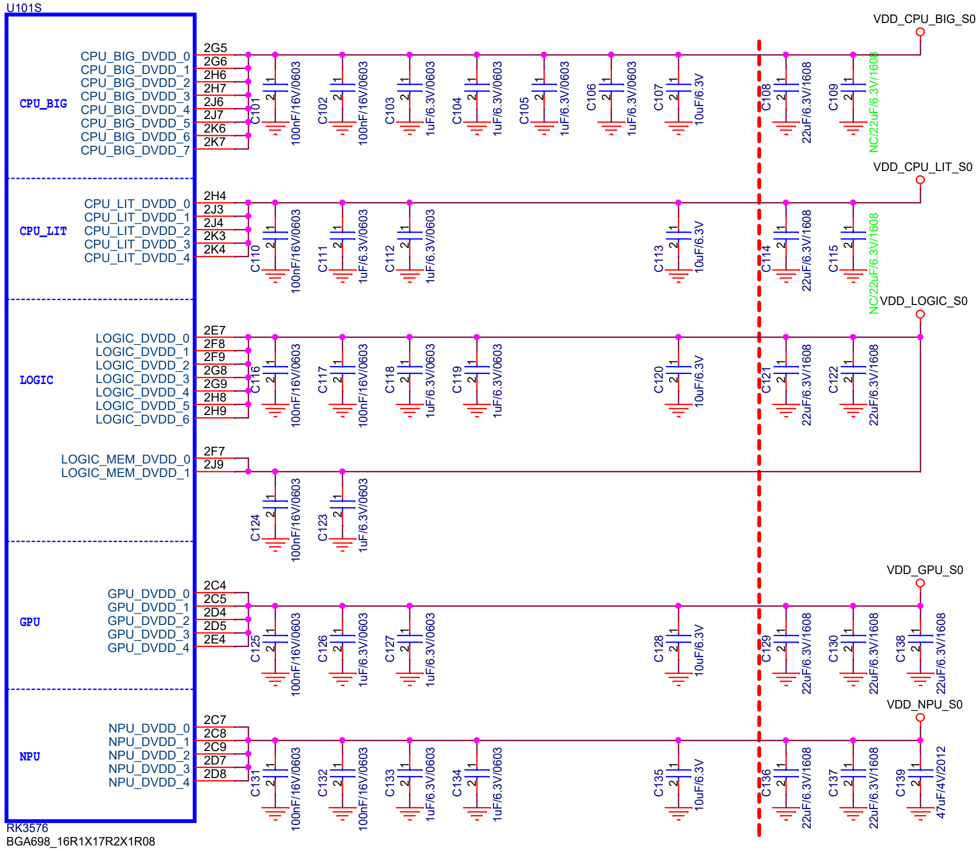
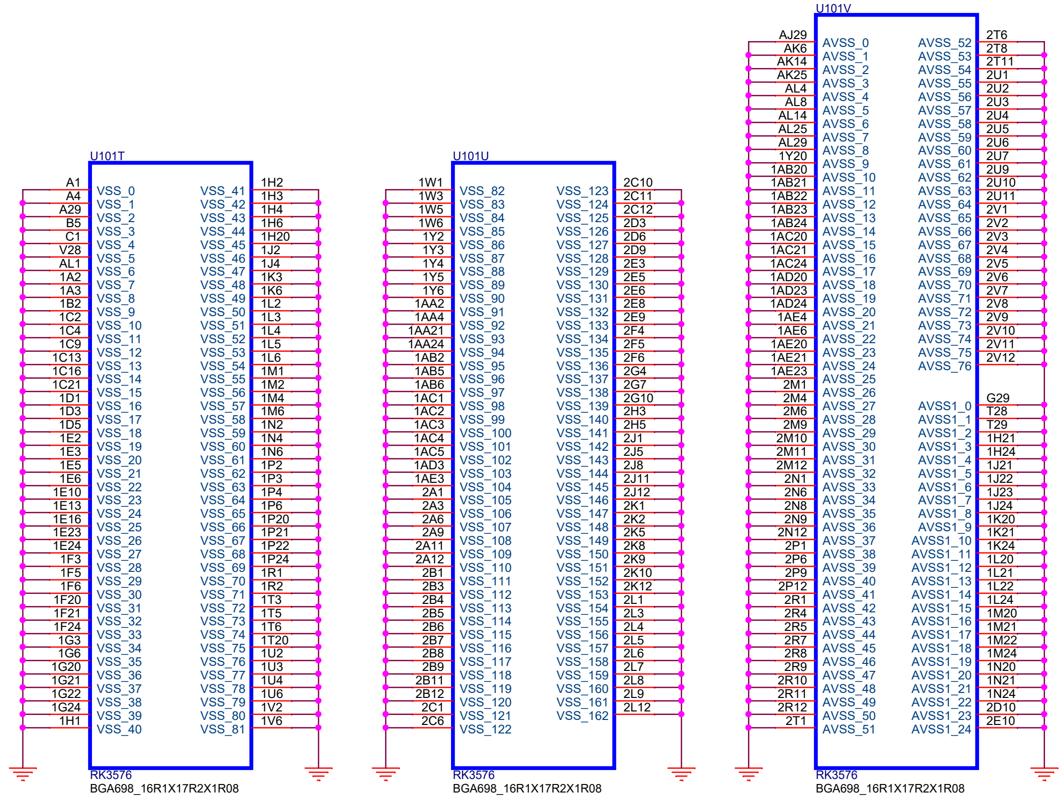
RK3576_S (Power)



Note:
Caps of between dashed red lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3576_T/U/V (GND)



Title	02.RK3576-Power/GND
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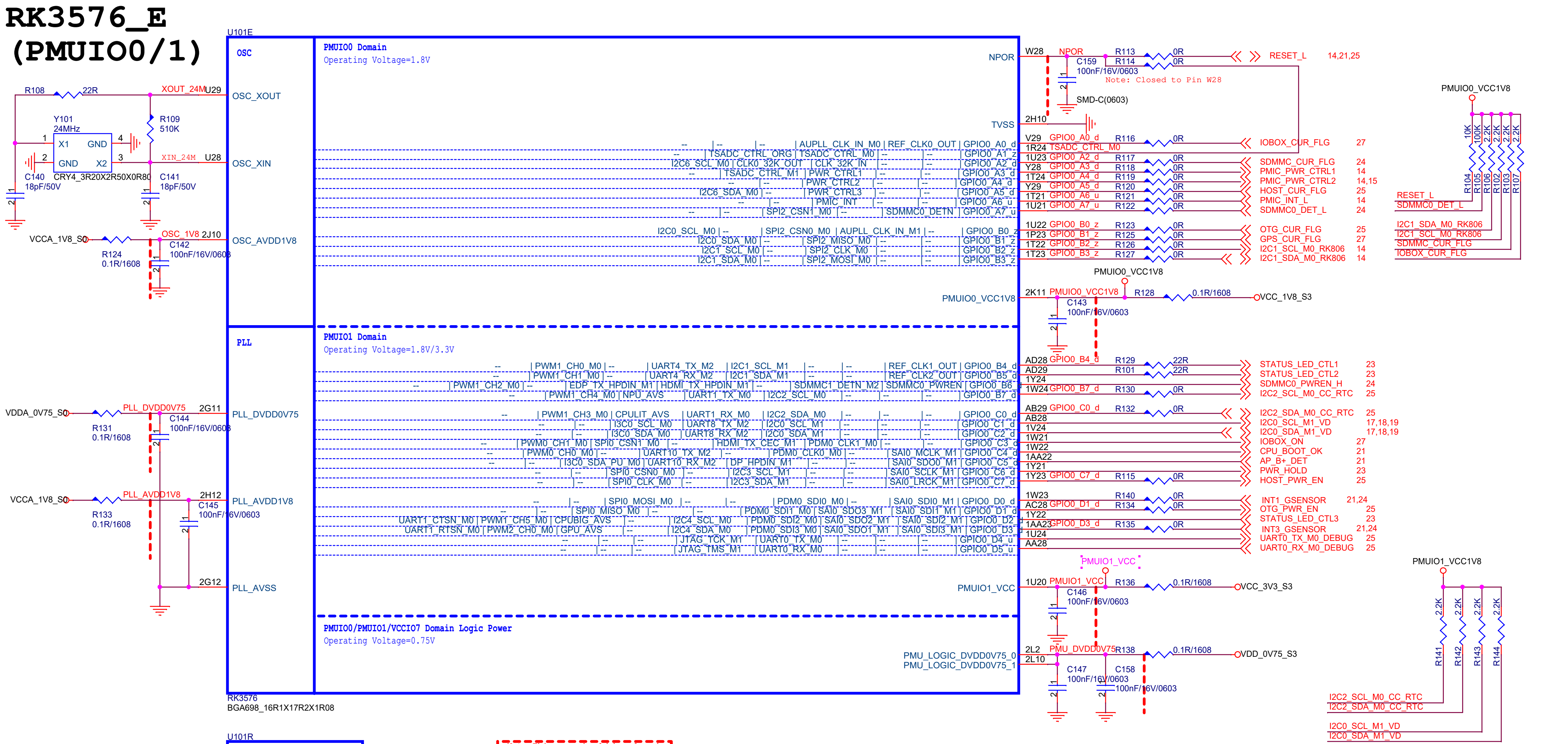
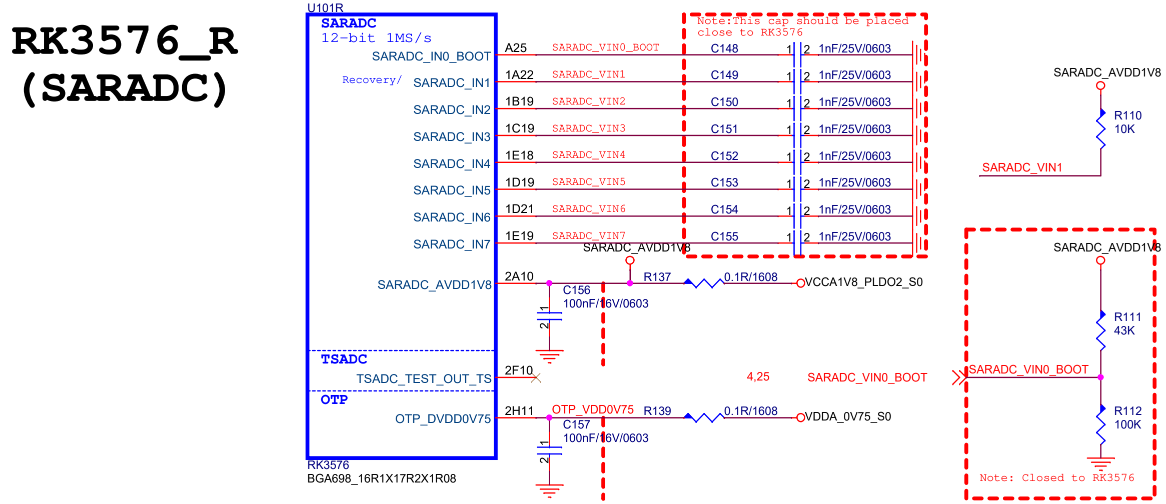


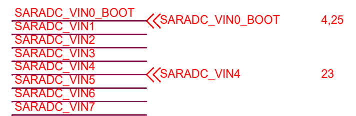


Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package.

BOOT MODE CONFIG

Config Table for SARADC_VIN0_BOOT

Item	Rup	Rdown	ADC Value	Boot Mode
Config1	NC	10K	0	USB (Maskrom mode)
Config2	100K	11.3K	416	FSPI0->USB
Config3	100K	24.9K	816	FSPI1_M0->EMMC->USB
Config4	100K	43K	1231	FSPI1_M1->EMMC->USB
Config5	100K	68K	1658	FSPI0->UFS->USB
Config6	100K	100K	2048	FSPI1_M0->UFS->USB
Config7	68K	100K	2438	UFS->USB
Config8	43K	100K	2864	UFS->SDMMC0->USB
Config9	24.9K	100K	3279	RFU
Config10	11.3K	100K	3679	EMMC->SDMMC0->USB
Config11	10K	NC	4095	EMMC->USB



Title: 03.RK3576-OSC/PLL/PMUIO/SARA
Size: A3
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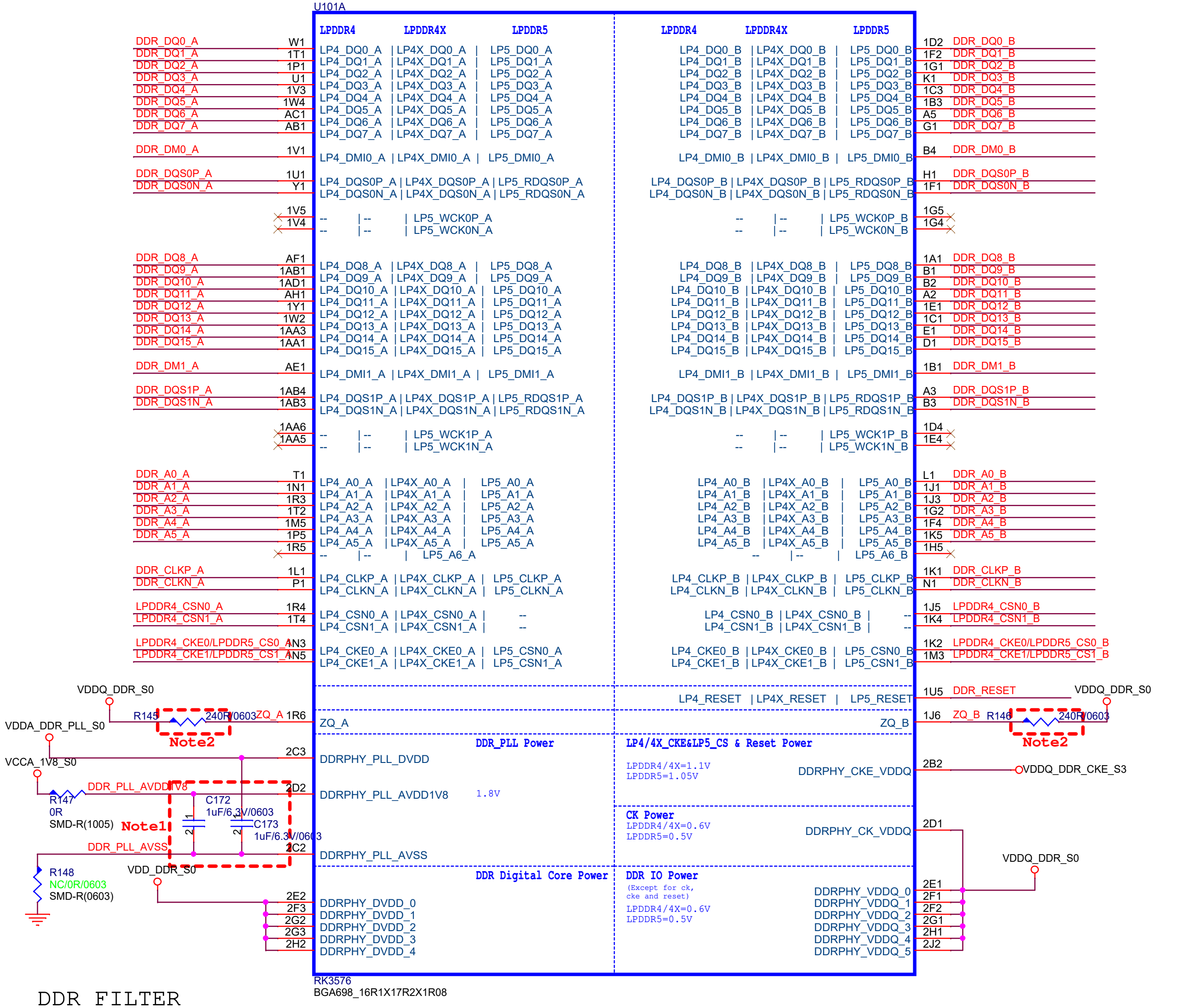
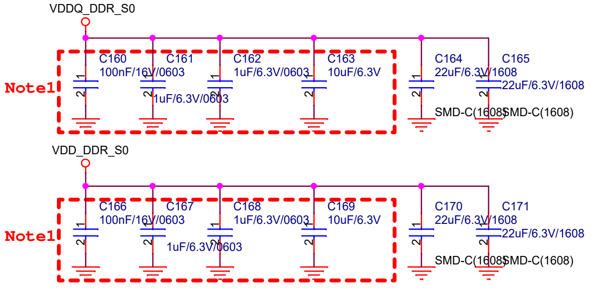
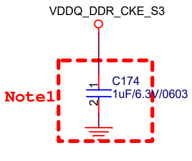
RK3576_A(DDRPHY)



DDR FILTER



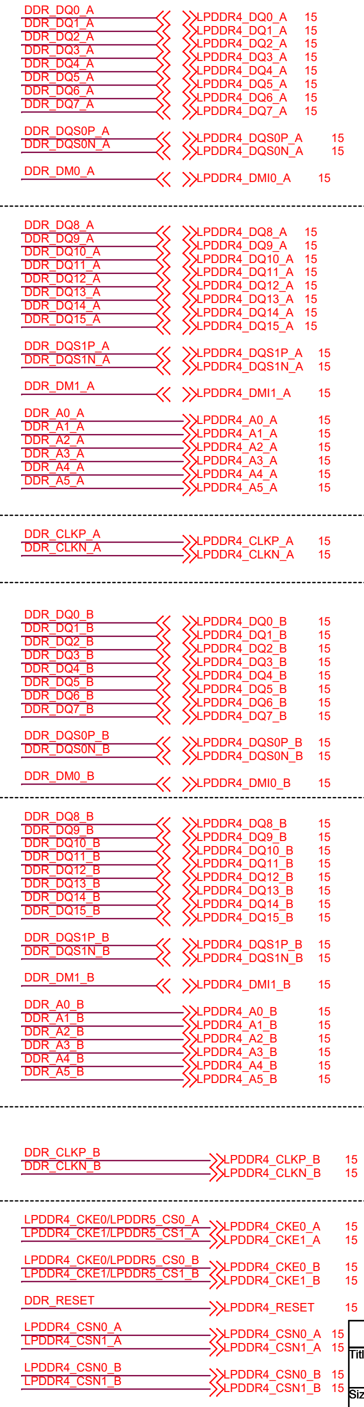


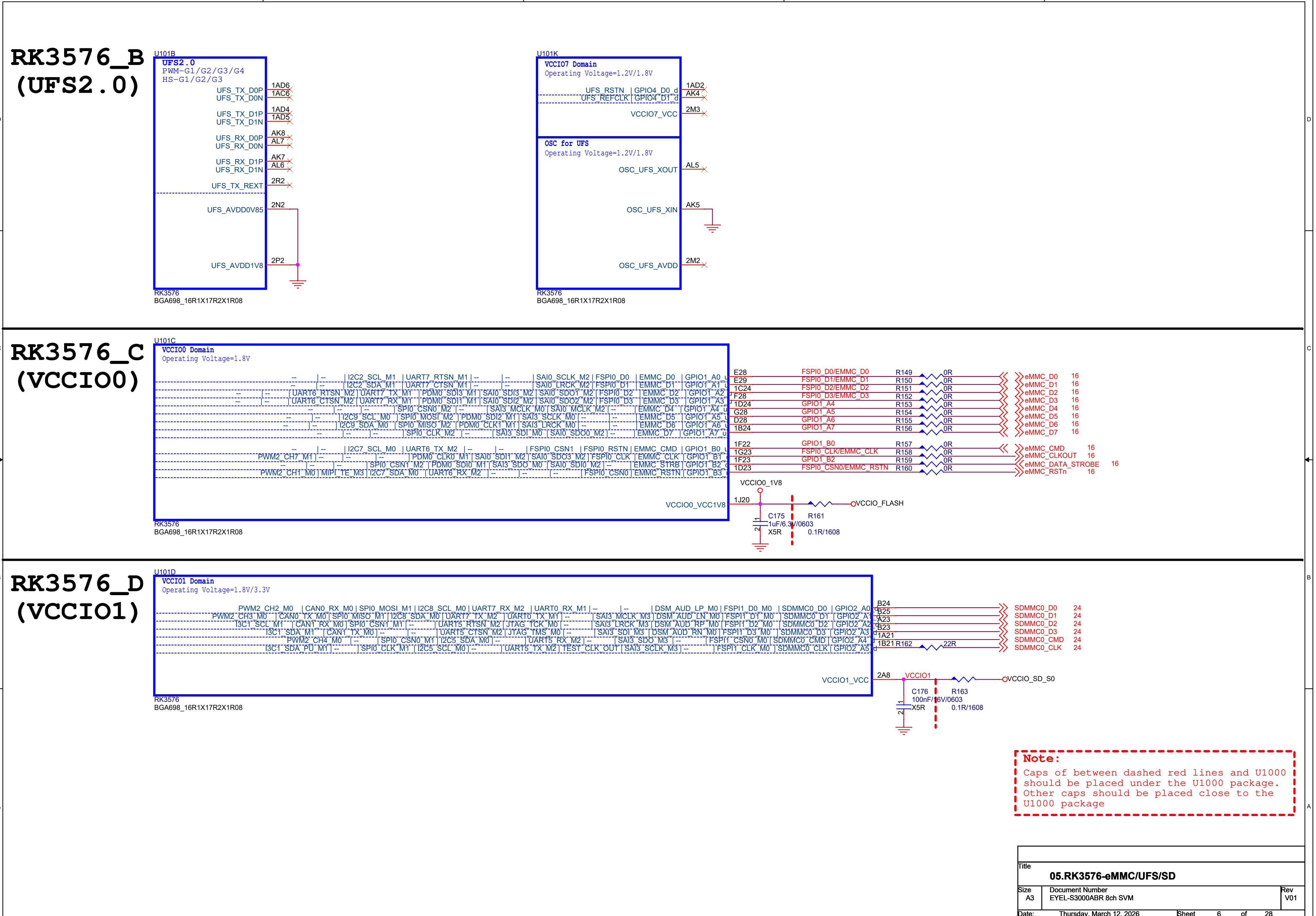
Note:
(1) Power Sequence: VDD-VDDQ_CKE-VDDQ
(2) Hold power of DDRPHY_CKE_VDDQ during retention times.

Note1:
Caps in the red line dotted box should be placed under the U1000 package

Note2:
Resistors in the red line dotted box should be placed under the U1000 package

LPDDR4 Signal





RK3576_B
(UFS2.0)
U101B
UFS2.0
PWM-G1/G2/G3/G4
HS-G1/G2/G3
UFS_TX_D0P 1AD6
UFS_TX_D0N 1AC6
UFS_TX_D1P 1AD4
UFS_TX_D1N 1AD5
UFS_RX_D0P AK8
UFS_RX_D0N AL7
UFS_RX_D1P AK7
UFS_RX_D1N AL6
UFS_TX_REXT 2R2
UFS_AVDD0V85 2N2
UFS_AVDD1V8 2P2
RK3576
BGA698_16R1X17R2X1R08
U101K
VCCIO7 Domain
Operating Voltage=1.2V/1.8V
UFS_RSTN | GPIO4_D0_d 1AD2
UFS_REFCLK | GPIO4_D1_d AK4
VCCIO7_VCC 2M3
OSC for UFS
Operating Voltage=1.2V/1.8V
OSC_UFS_XOUT AL5
OSC_UFS_XIN AK5
OSC_UFS_AVDD 2M2
RK3576
BGA698_16R1X17R2X1R08
RK3576_C
(VCCIO0)
U101C
VCCIO0 Domain
Operating Voltage=1.8V
E28 FSPI0_D0/EMMC_D0 R149 0R eMMC_D0 16
E29 FSPI0_D1/EMMC_D1 R150 0R eMMC_D1 16
1C24 FSPI0_D2/EMMC_D2 R151 0R eMMC_D2 16
F28 FSPI0_D3/EMMC_D3 R152 0R eMMC_D3 16
1D24 GPIO1_A4 R153 0R eMMC_D4 16
G28 GPIO1_A5 R154 0R eMMC_D5 16
D28 GPIO1_A6 R155 0R eMMC_D6 16
1B24 GPIO1_A7 R156 0R eMMC_D7 16
1F22 GPIO1_B0 R157 0R eMMC_CMD 16
1G23 FSPI0_CLK/EMMC_CLK R158 0R eMMC_CLKOUT 16
1F23 GPIO1_B2 R159 0R eMMC_DATA_STROBE 16
1D23 FSPI0_CSN0/EMMC_RSTN R160 0R eMMC_RSTn 16
VCCIO0_1V8
VCCIO0_VCC1V8 1J20
C175 1uF/6.3V/0603 X5R
R161 0.1R/1608
VCCIO_FLASH
RK3576
BGA698_16R1X17R2X1R08
RK3576_D
(VCCIO1)
U101D
VCCIO1 Domain
Operating Voltage=1.8V/3.3V
B24 SDMMC0_D0 24
B25 SDMMC0_D1 24
A23 SDMMC0_D2 24
B23 SDMMC0_D3 24
1A21 SDMMC0_CMD 24
1B21 R162 22R SDMMC0_CLK 24
VCCIO1_VCC 2A8 VCCIO1
C176 100nF/16V/0603 X5R
R163 0.1R/1608
VCCIO_SD_S0
RK3576
BGA698_16R1X17R2X1R08
Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package
Title
05.RK3576-eMMC/UFS/SD
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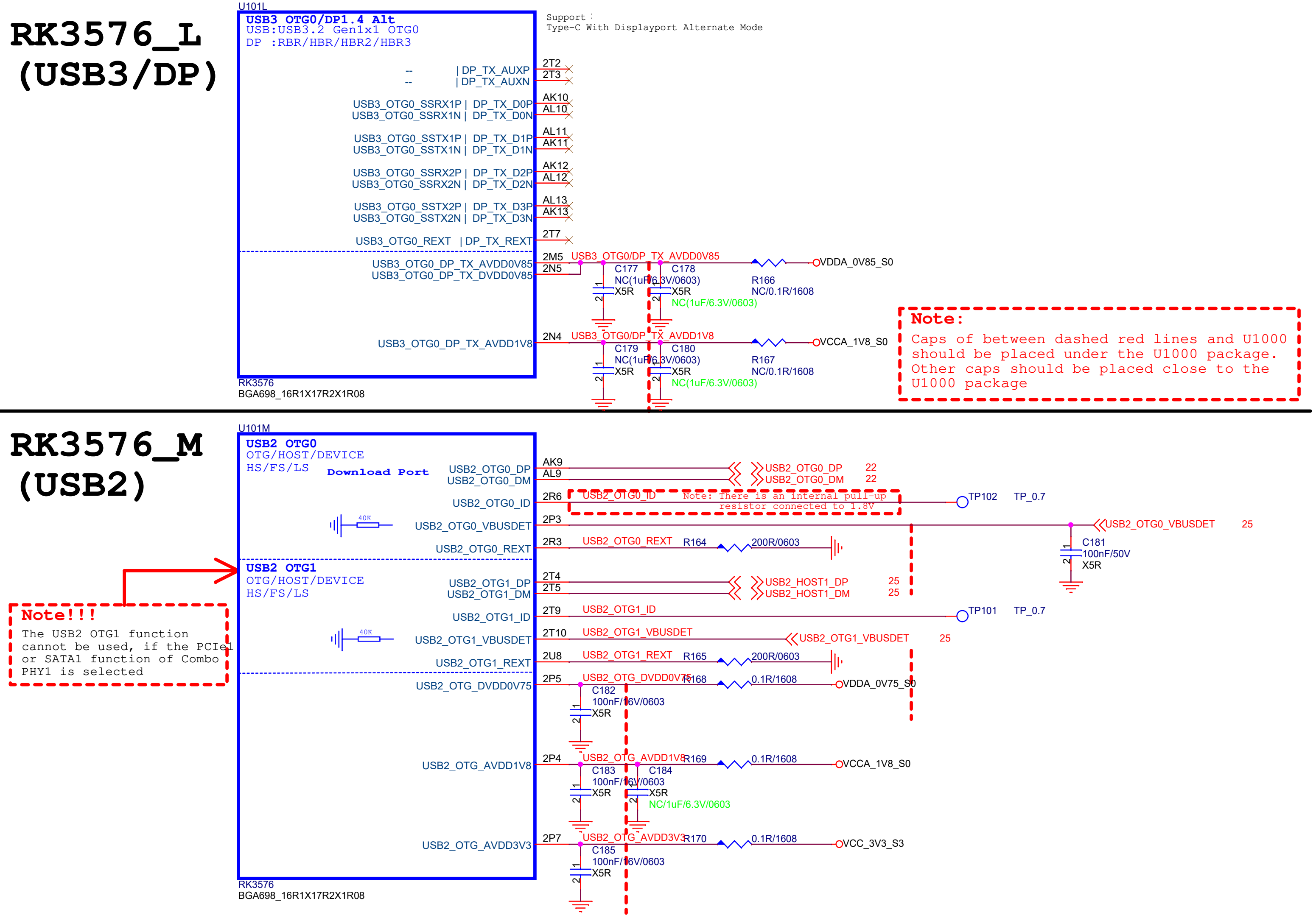

Title	06.RK3576-TypeC/USB	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
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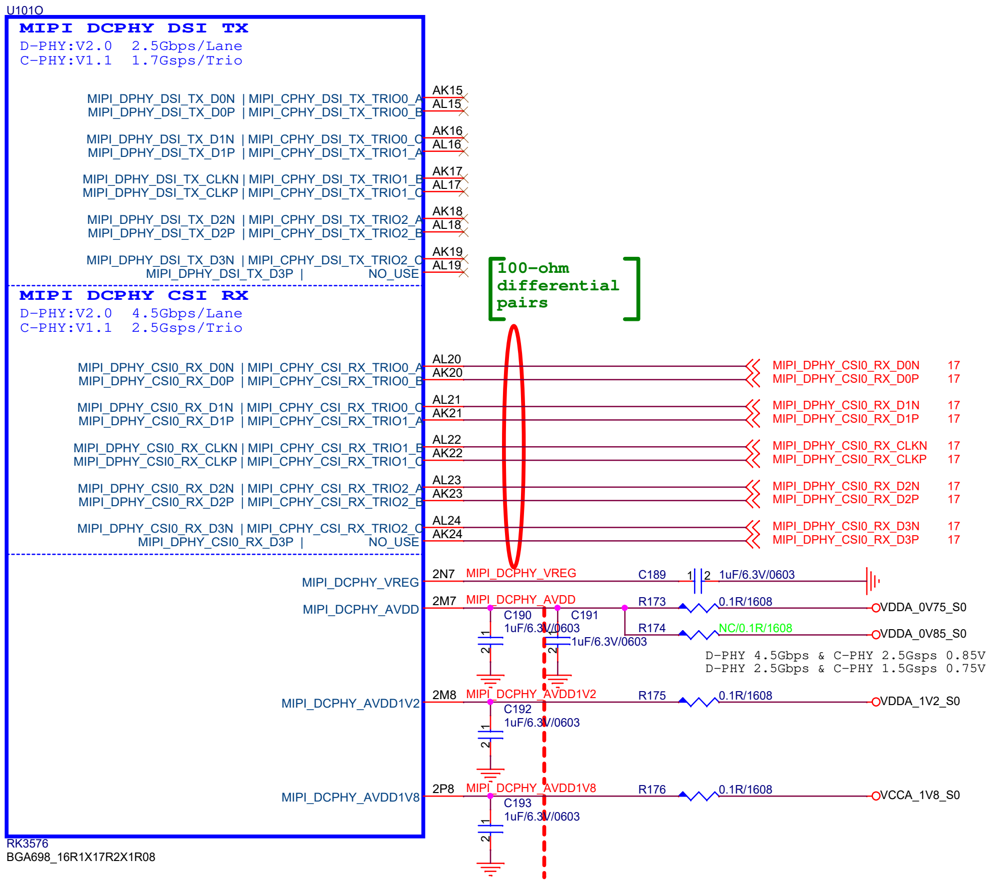
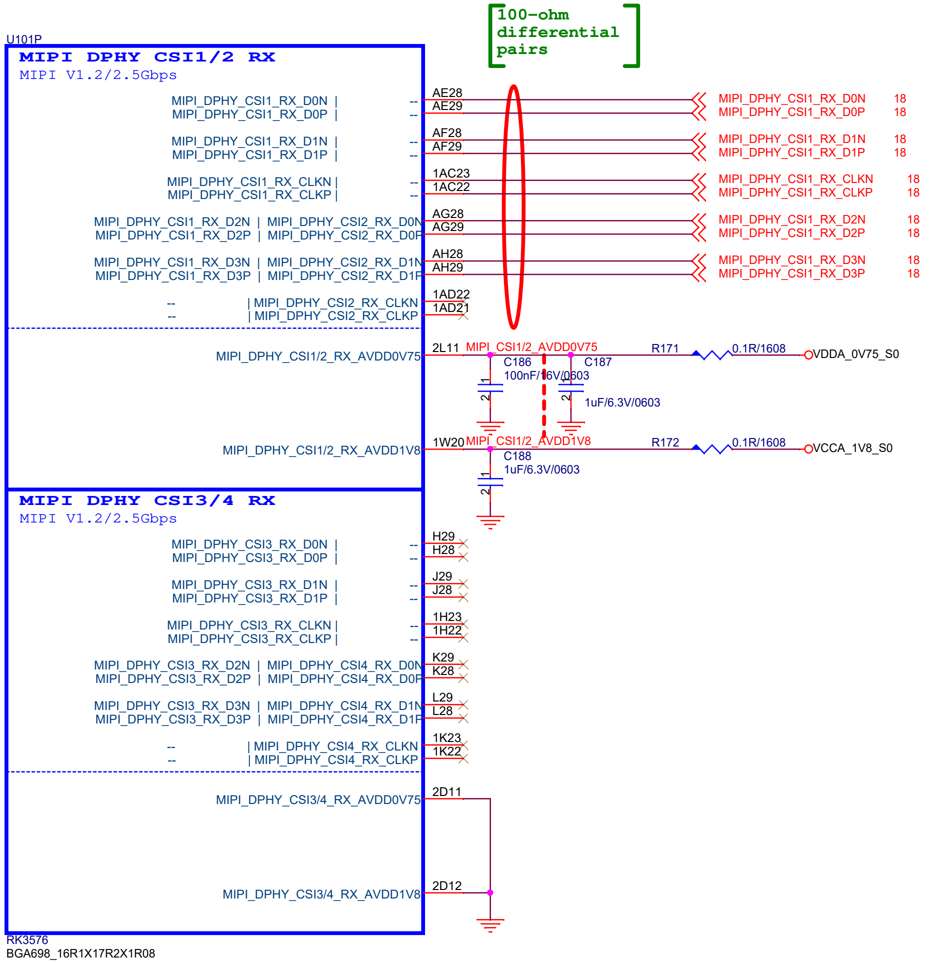
RK3576_O(MIPI DCPHY)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

RK3576_P(MIPI DPHY CSI RX)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

Title	07.RK3576-MIPI DSI/CSI
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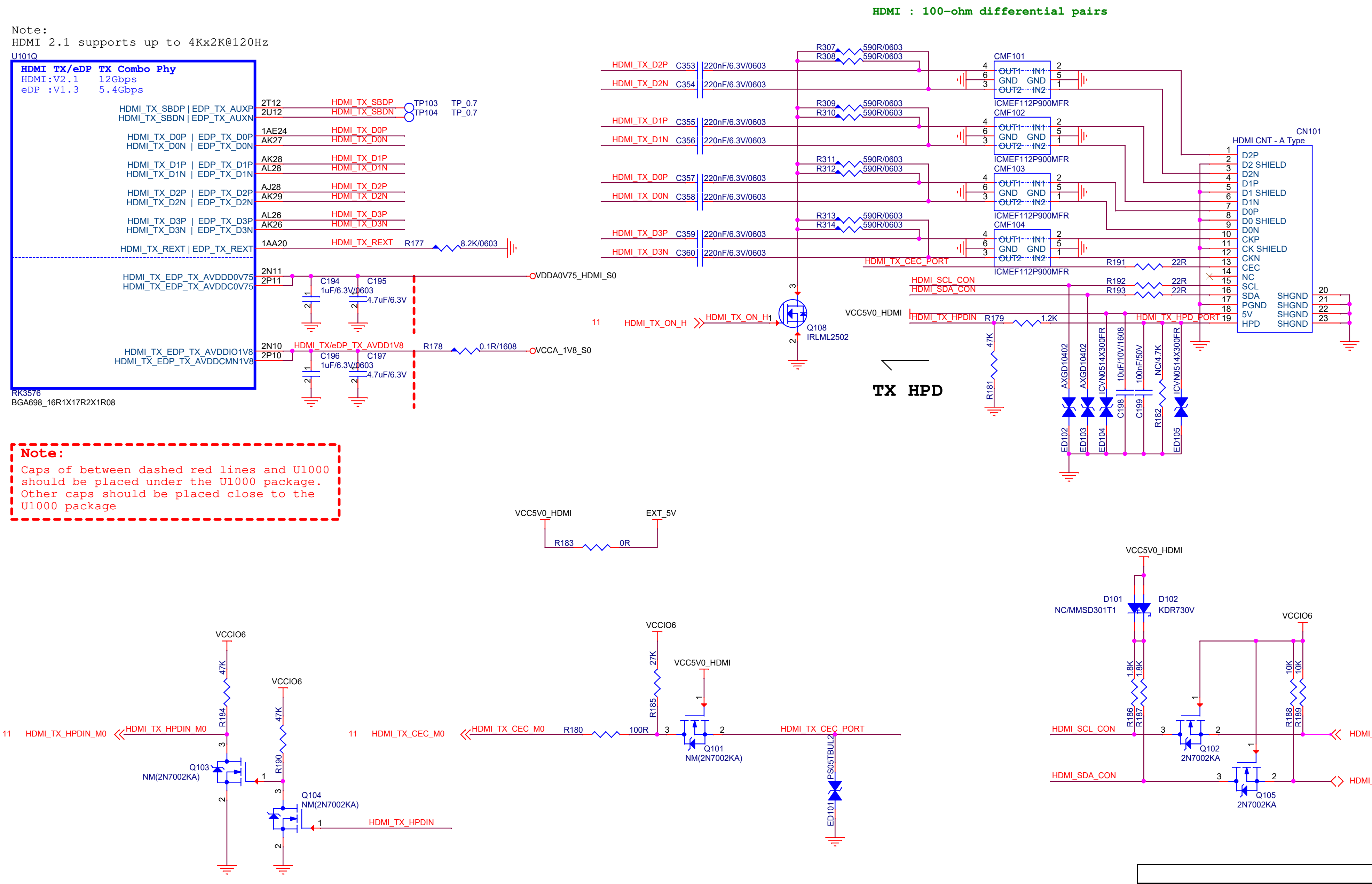
RK3576_Q(HDMI/eDP)

Note:
HDMI 2.1 supports up to 4Kx2K@120Hz



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

Title	08.RK3576-HDMI/eDP
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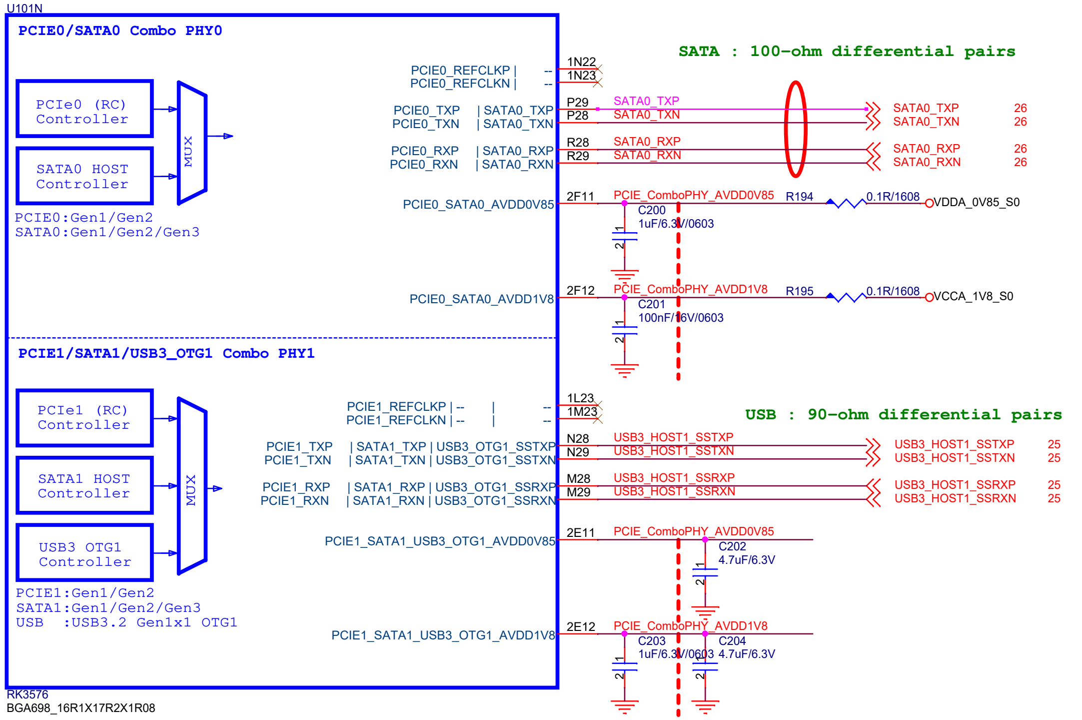
RK3576_N(PCIe/SATA/USB3)

PCIe Port0



Note!!!

If the PCIe1 or SATA1 function of Combo PHY1 is selected, the USB3 OTG1 function cannot be used, and even the USB2 OTG1 function cannot be used

Note:

Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

Title	09.RK3576-PCIe/SATA/USB3
Size	A3
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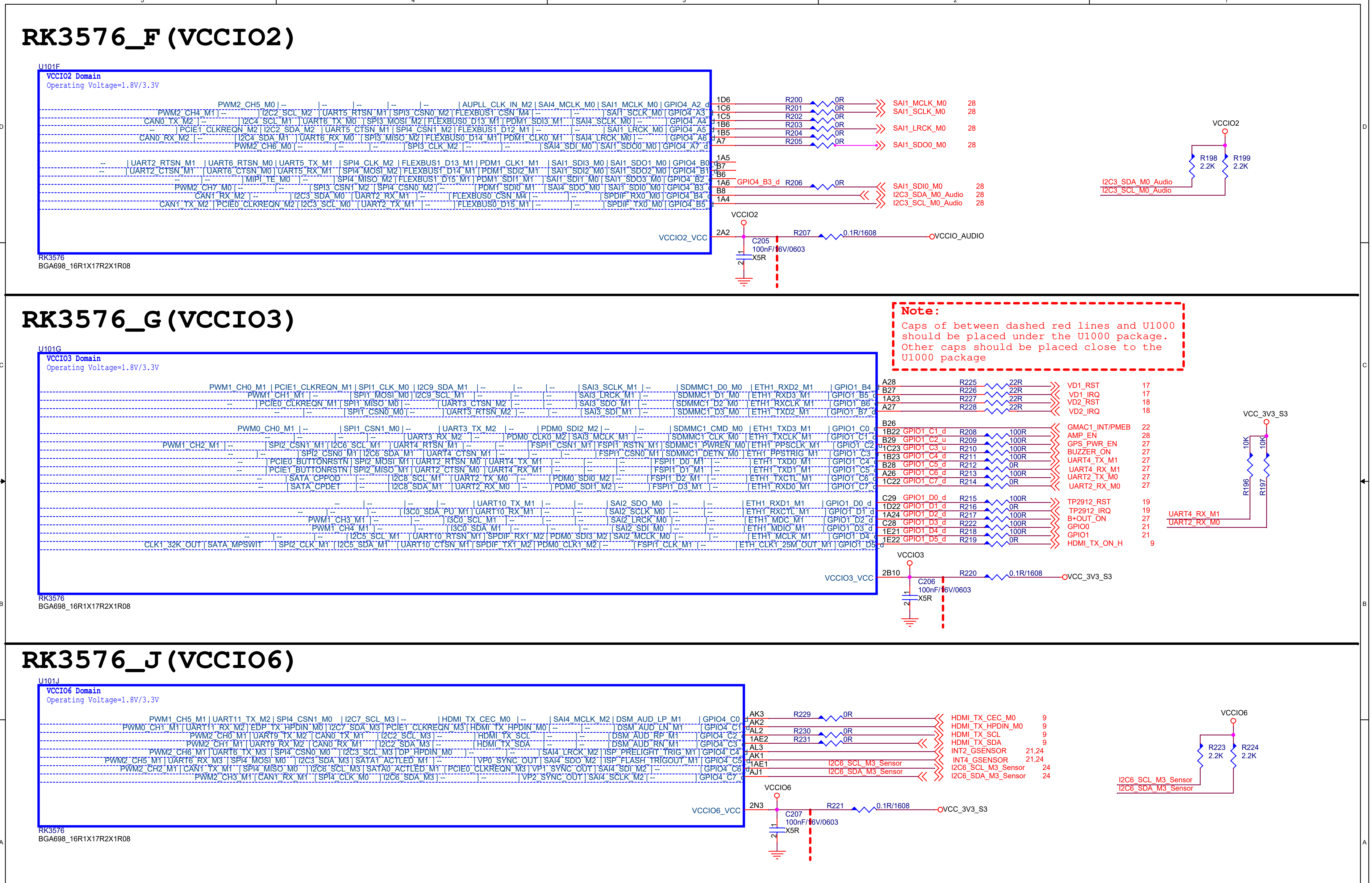

Title	10.RK3576-GPIO VCCIO2/3/6	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
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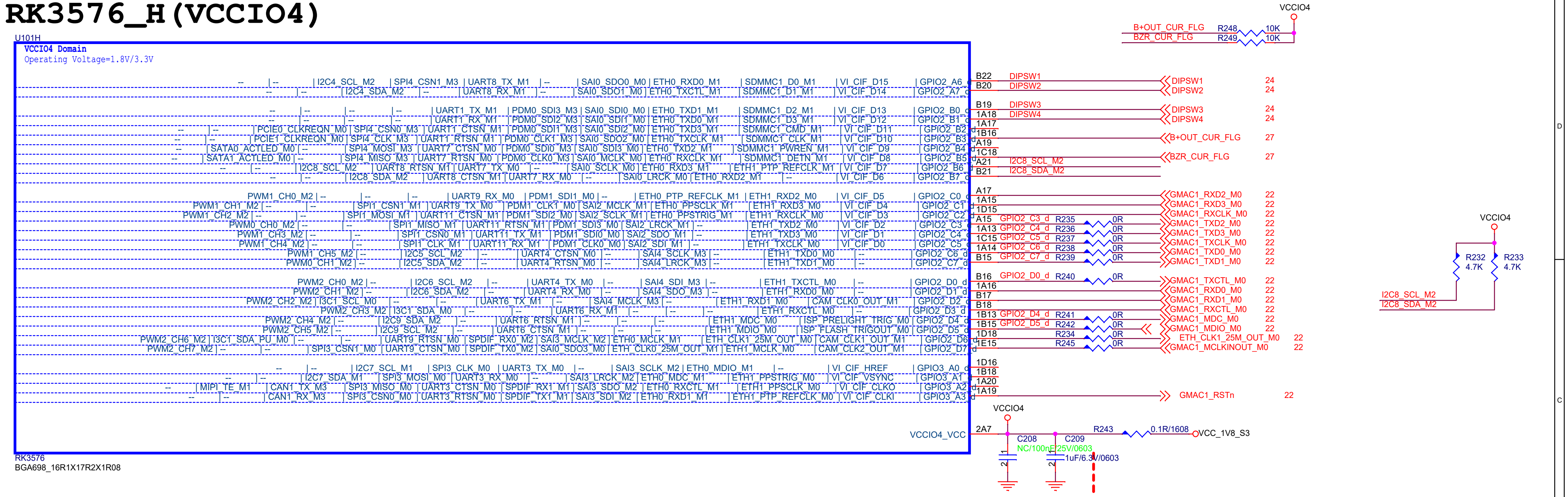
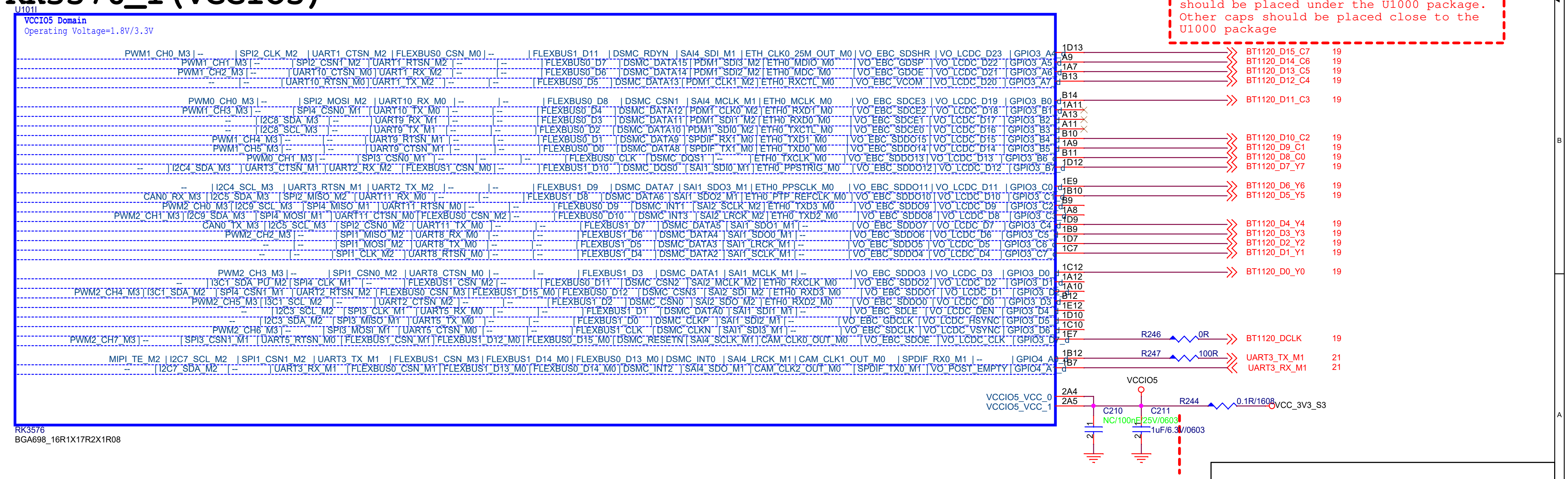
RK3576_H(VCCIO4)



RK3576_I(VCCIO5)

Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package



Title	11.RK3576-GPIO VCCIO4/5	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
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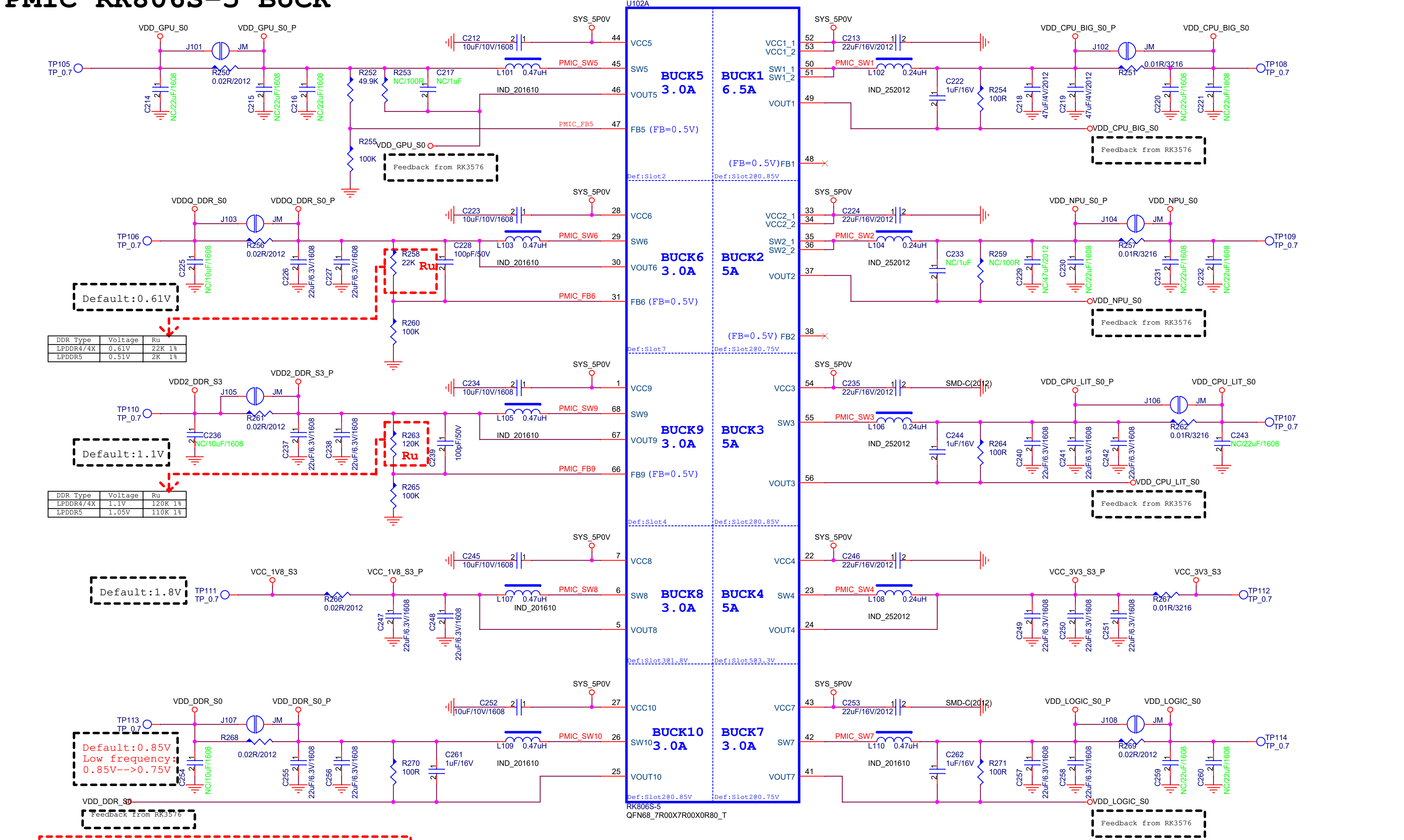
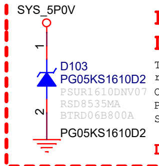
PMIC RK806S-5 BUCK



DDR Type	Voltage	Ru
LPDDR4/4X	0.61V	22K 1%
LPDDR5	0.51V	2K 1%

DDR Type	Voltage	Ru
LPDDR4/4X	1.1V	120K 1%
LPDDR5	1.05V	110K 1%

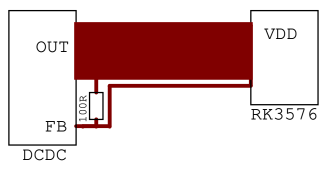


IF TVS UNMOUNTED,
ESD OR SURGE SHOULD BE
DAMAGE THE PMIC!!!

This device must be mounted.Replacing TVS mode is not recommended, if must, please choose the same specifications
Operating Supply Vltage : 5.5V(5.25-6V)
PeakPulse Current : >10A (tp=8/20uS)
Surge Clamping Voltage : <6.5V

DO NOT DELETE IT!

4,14 I2C1_SDA_M0_RK806
4,14 I2C1_SCL_M0_RK806
4,14 PMIC_PWR_CTRL1
4,14,15 PMIC_PWR_CTRL2
4,14 PMIC_INT_L
4,14,21,25 RESET_L
14,21 PWRON_L



Title	12.Power-PMIC RK806S-5_BUCK
Size	A3
Document Number	EYEL-S3000ABR 8ch SVM
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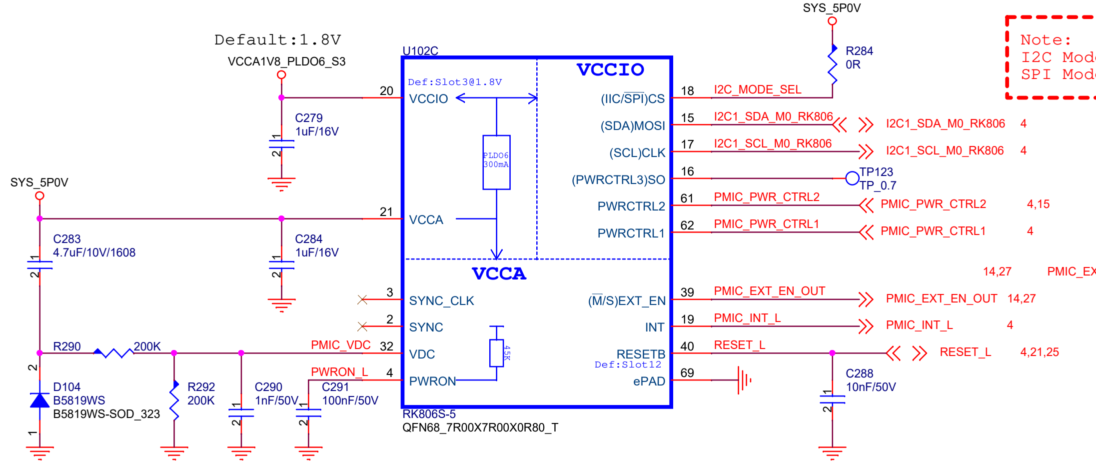
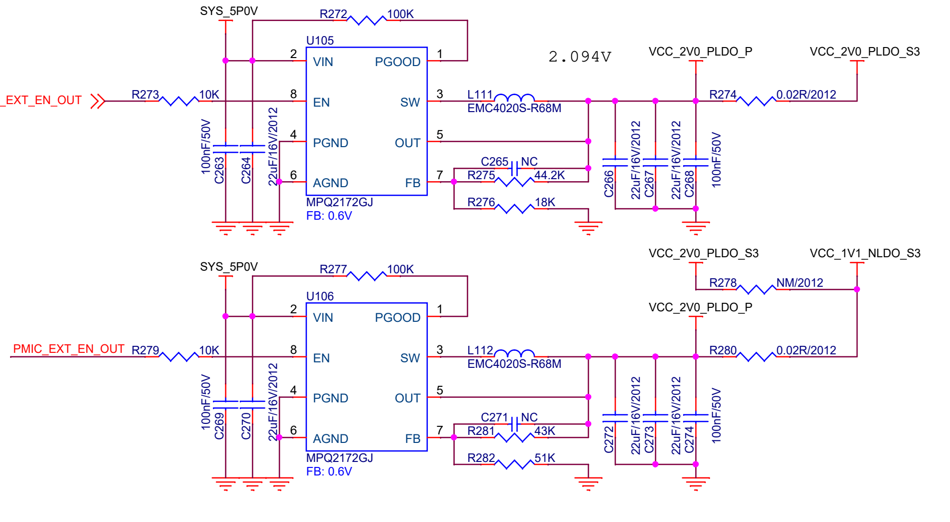
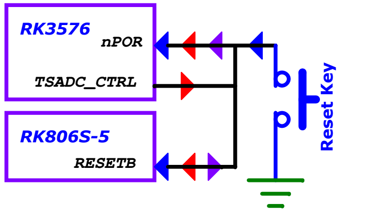
PMIC RK806S-5 Managerment



Note:
I2C Mode:CS(pin18) connected to VCCA(pin21);
SPI Mode(Def):CS(pin18) floating or connected to GND

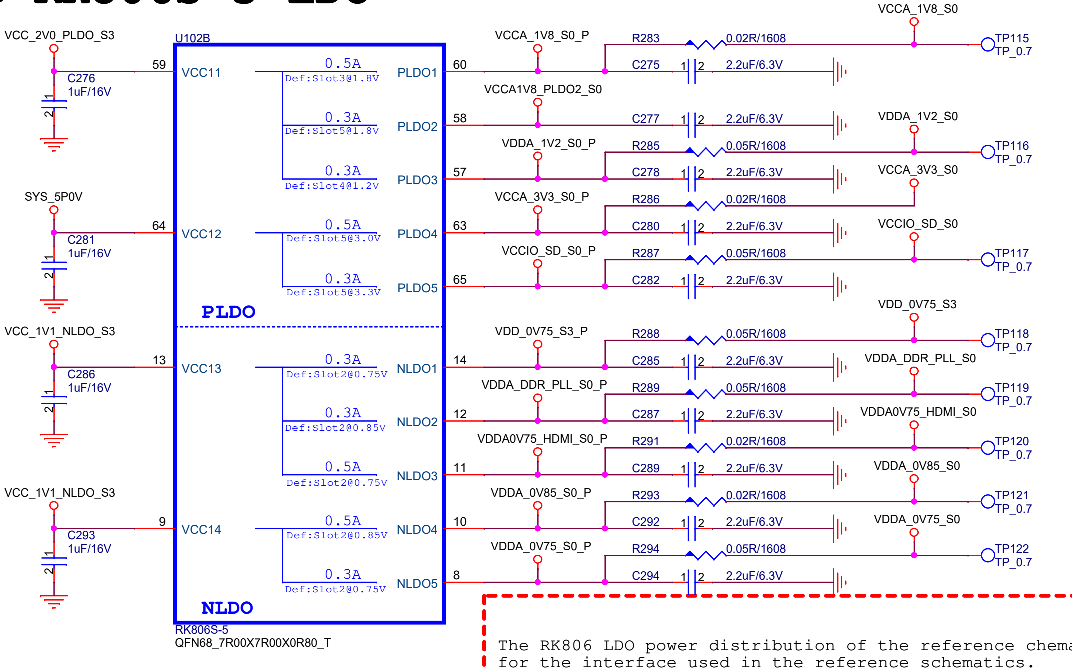




Note:

▶	Reset Key Control Path
▶	TSADC_SHUT Control Path
▶	RK806S-5 Control Path

PMIC RK806S-5 LDO



Default:1.8V

Default:1.8V

Default:1.2V

Default:3.0V

Default:3.3V

Default:0.75V

Default:0.85V;Low frequency:0.85V-->0.75V

Default:0.75V

Default:0.85V

Default:0.75V

The RK806 LDO power distribution of the reference chematics is only suitable for the interface used in the reference schematics.
If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

Title	13.Power-PMIC RK806S-5 LDO		
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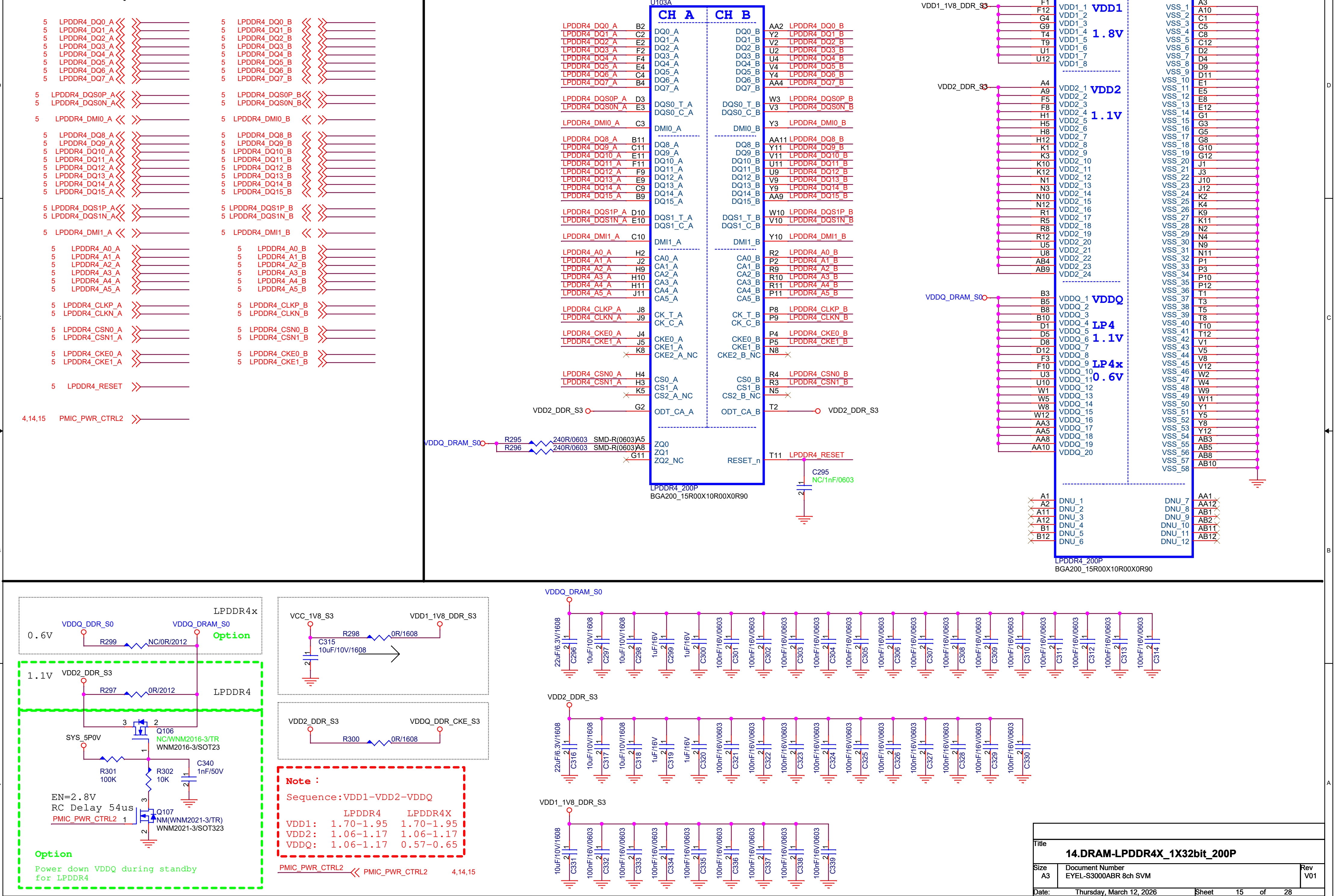
LPDDR4/4X



Title	14.DRAM-LPDDR4X_1X32bit_200P		
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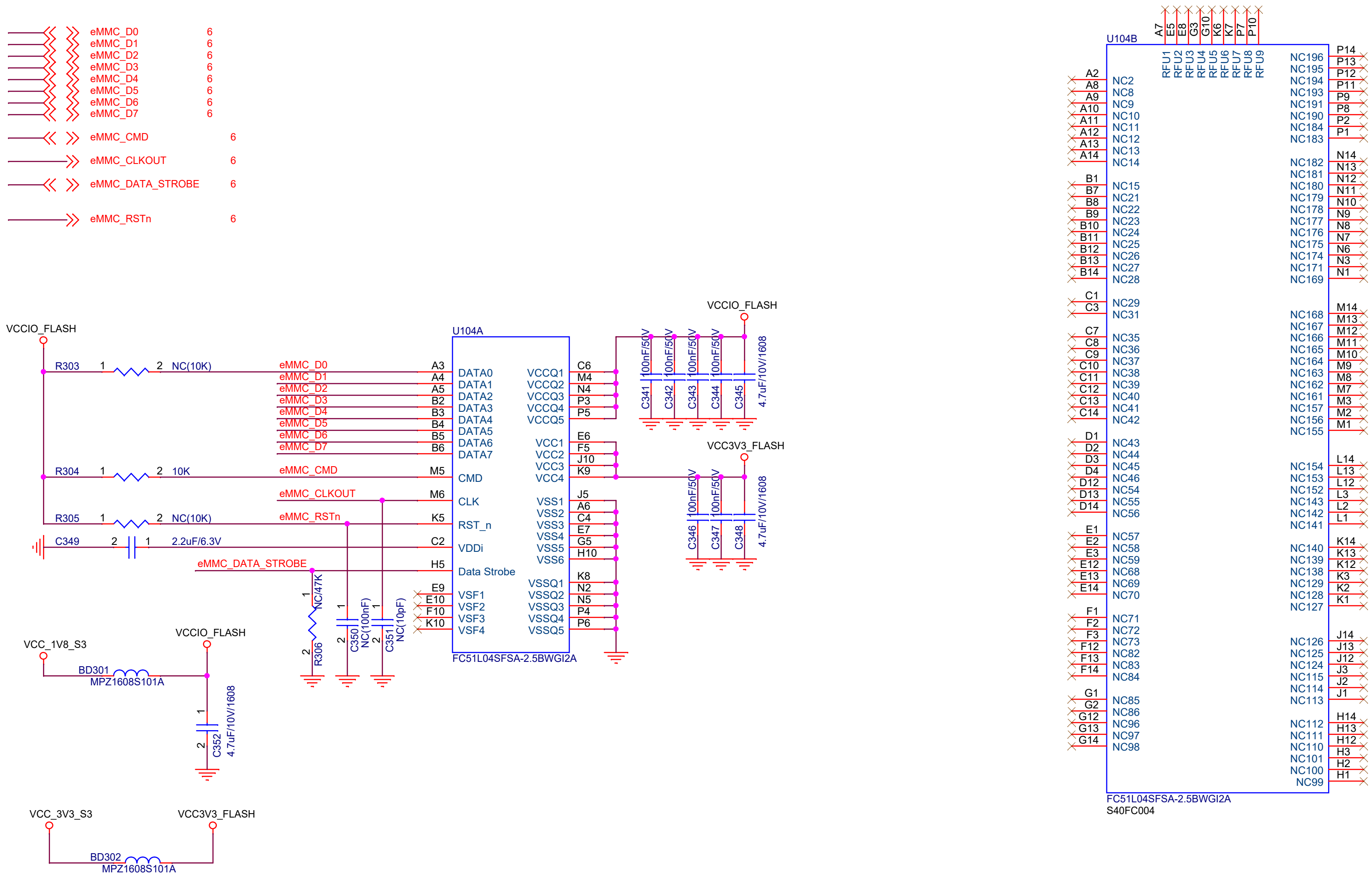

Title	15.eMMC	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
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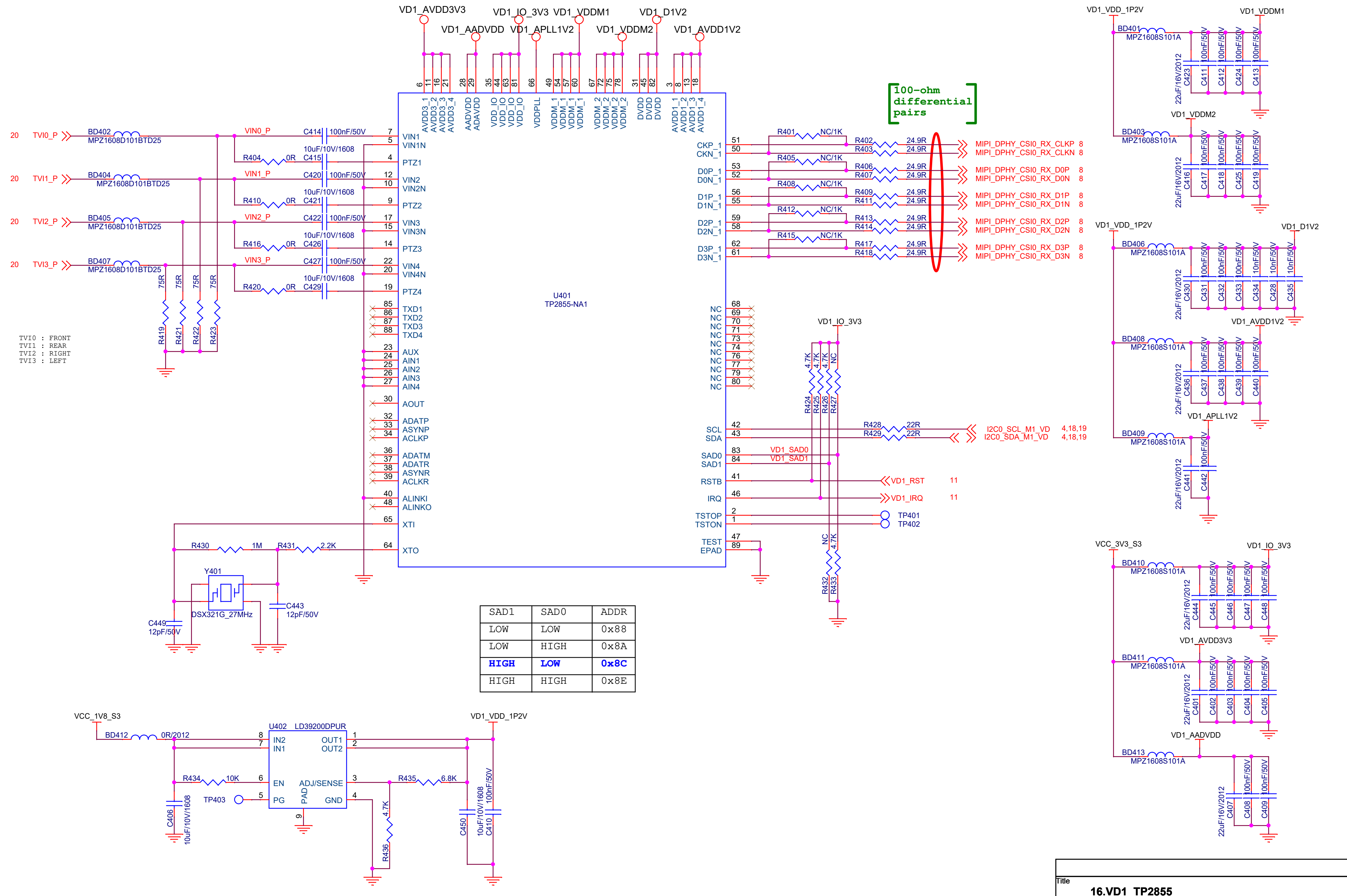
VIDEO DECODER1



SAD1	SAD0	ADDR
LOW	LOW	0x88
LOW	HIGH	0x8A
HIGH	**LOW**	**0x8C**
HIGH	HIGH	0x8E

Title
16.VD1_TP2855

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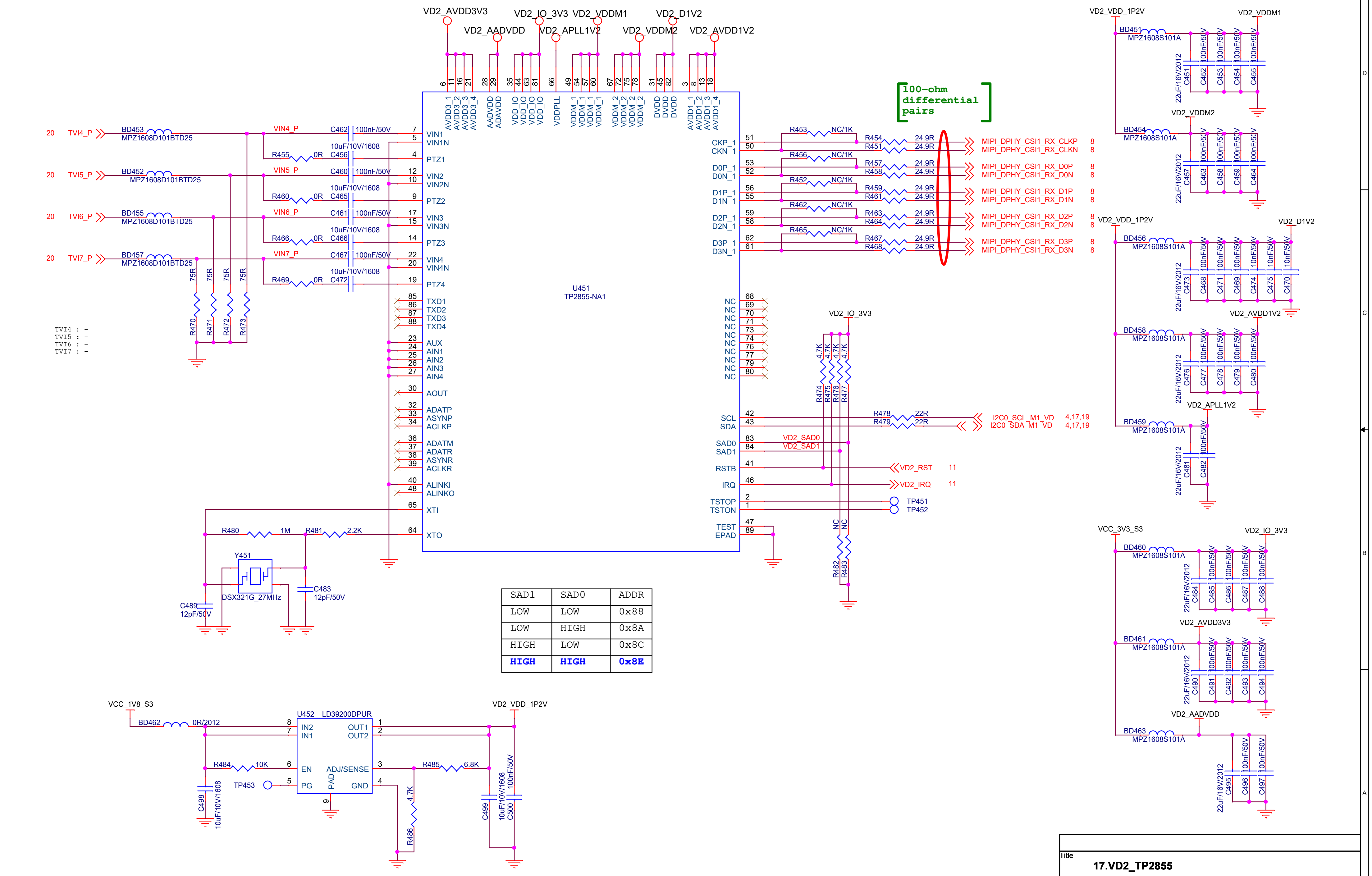
VIDEO DECODER2



SAD1	SAD0	ADDR
LOW	LOW	0x88
LOW	HIGH	0x8A
HIGH	LOW	0x8C
HIGH	**HIGH**	**0x8E**

Title	17.VD2_TP2855
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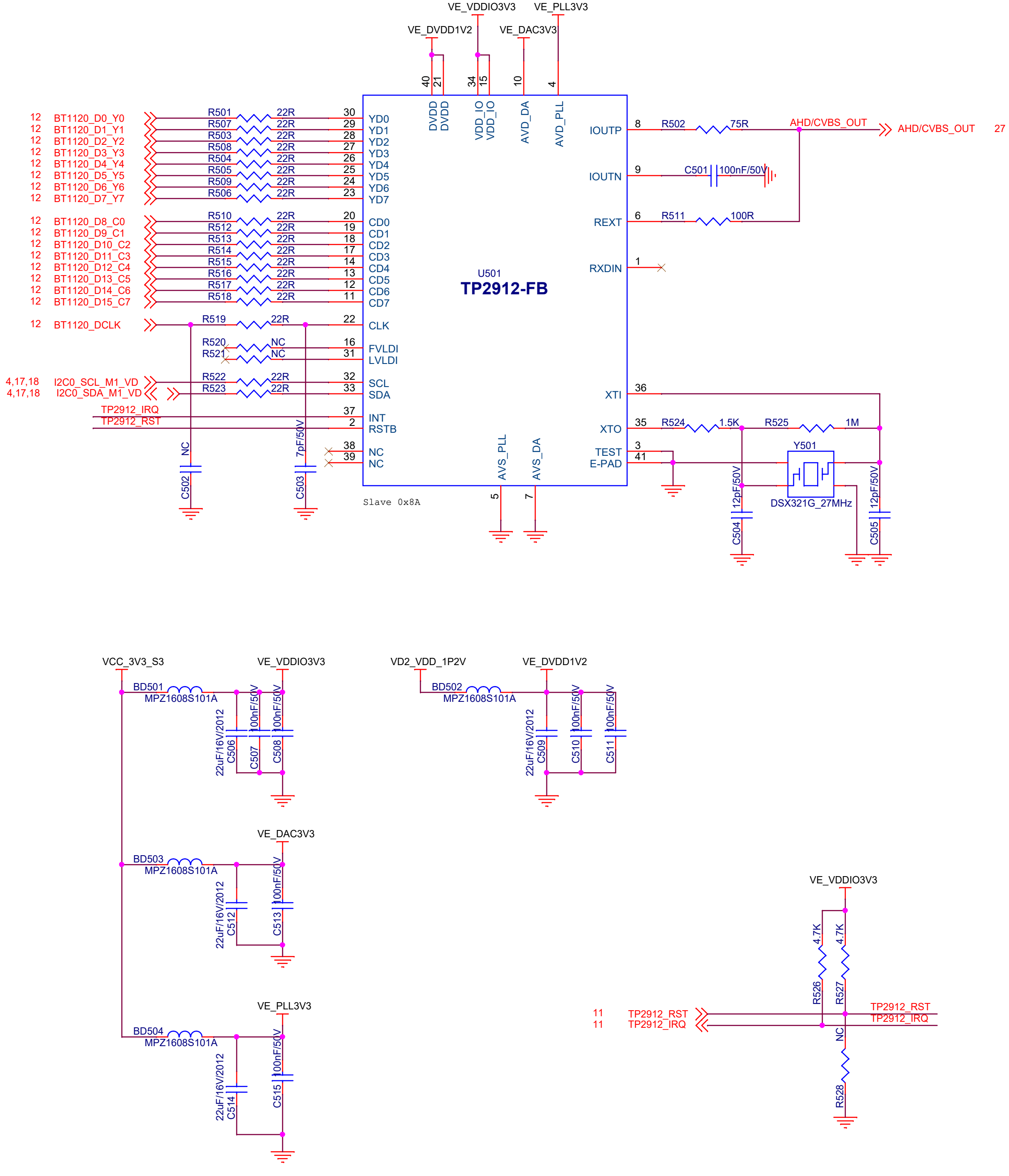
VIDEO ENCODER(AHD)



Title	18.VIDEO ENCODER	
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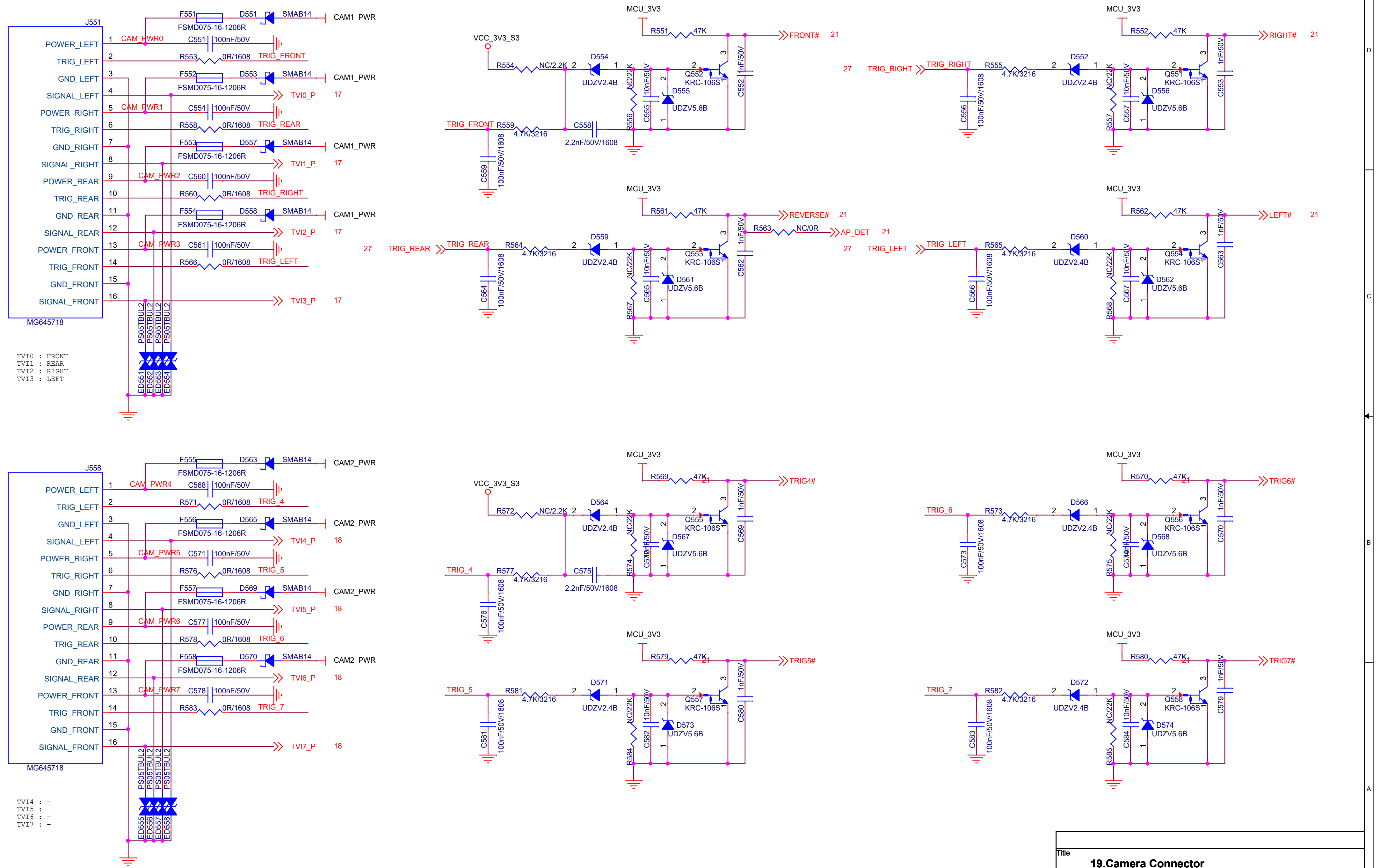
Camera Connector



Title	19.Camera Connector	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
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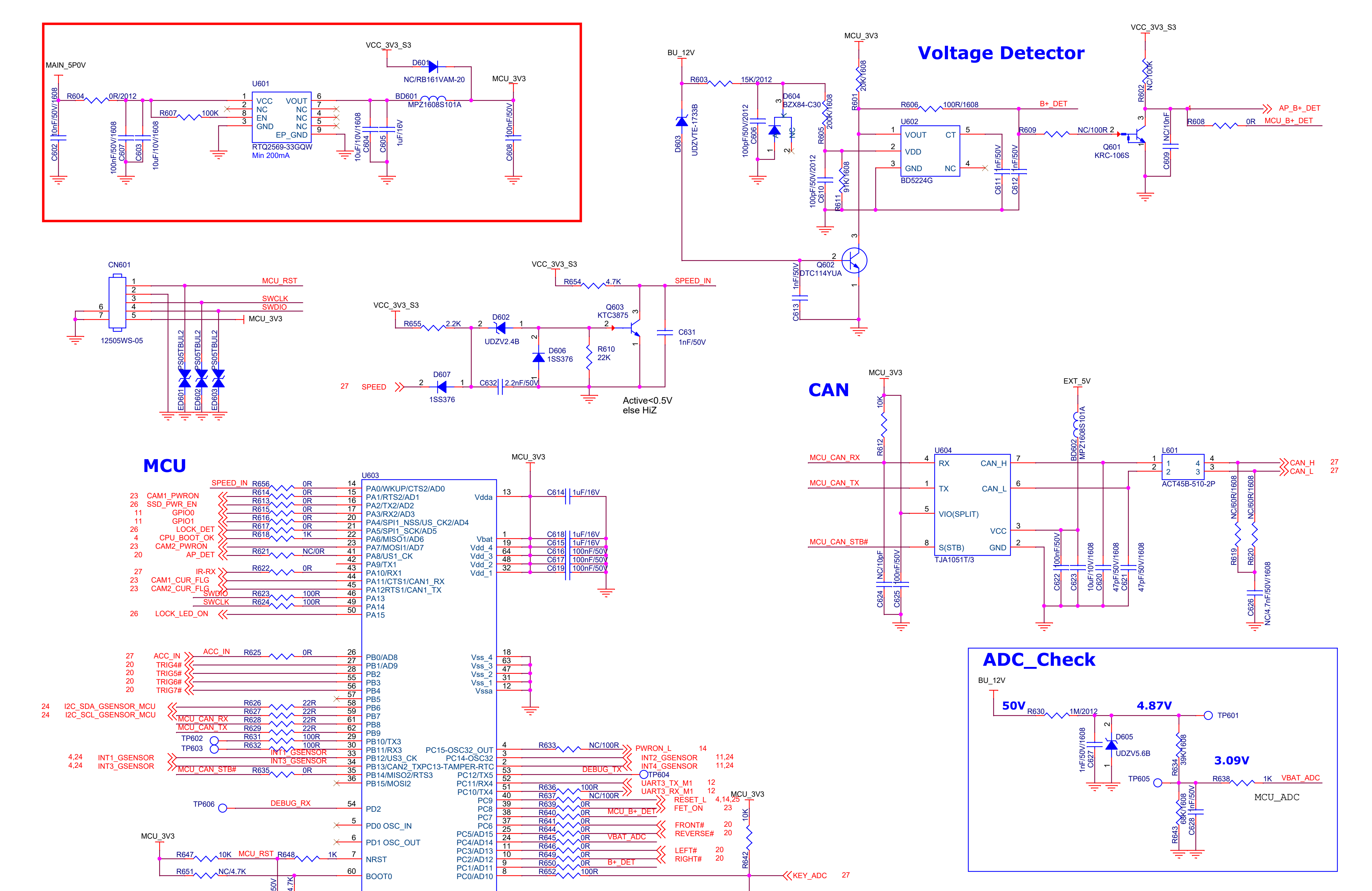

Title	20.MCU/CAN
Size	A3
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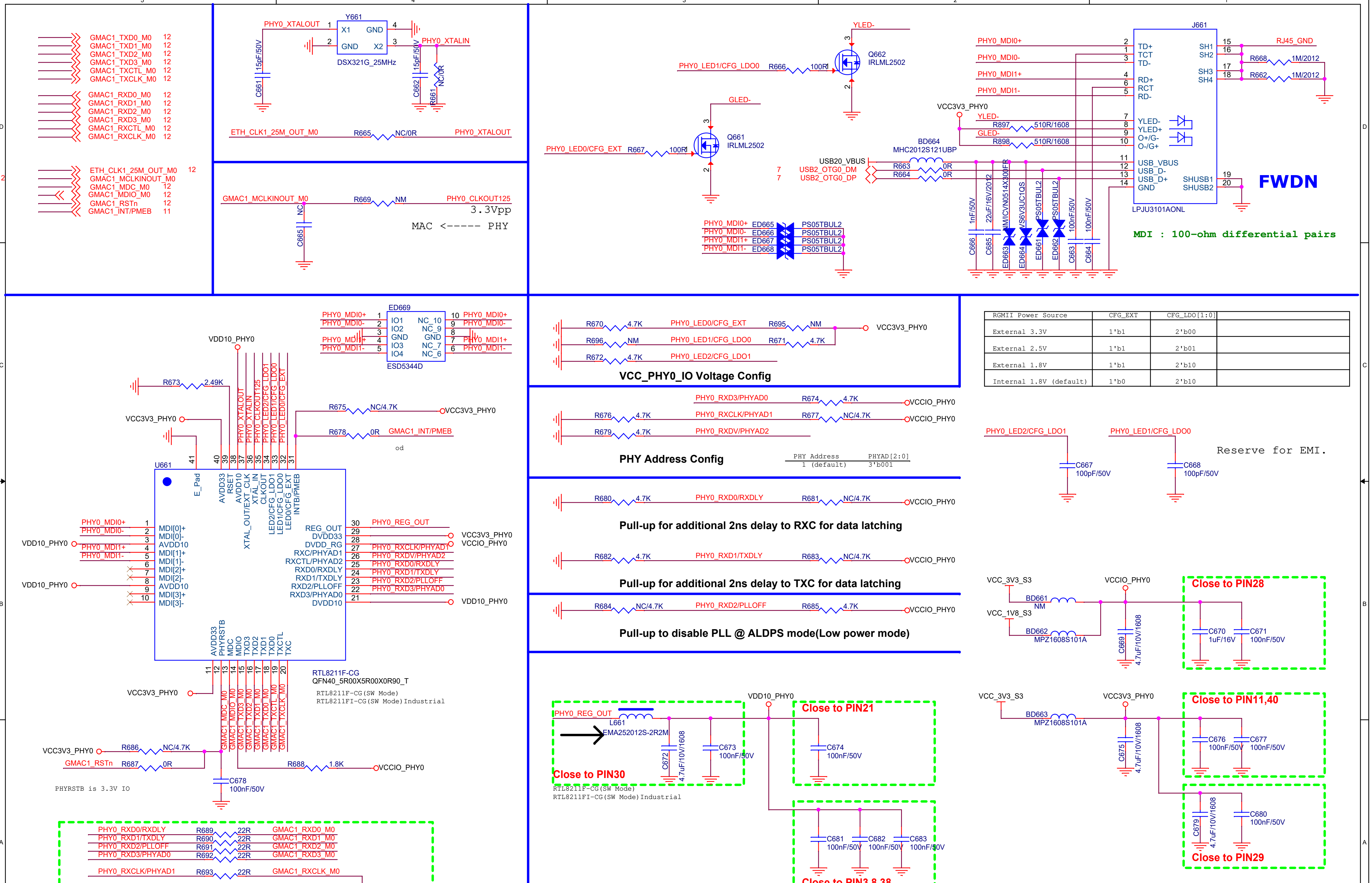

RGMII Power Source	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 2.5V	1'b1	2'b01
External 1.8V	1'b1	2'b10
Internal 1.8V (default)	1'b0	2'b10

PHY Address	PHYAD[2:0]
1 (default)	3'b001

Title: **21.ETHERNET**

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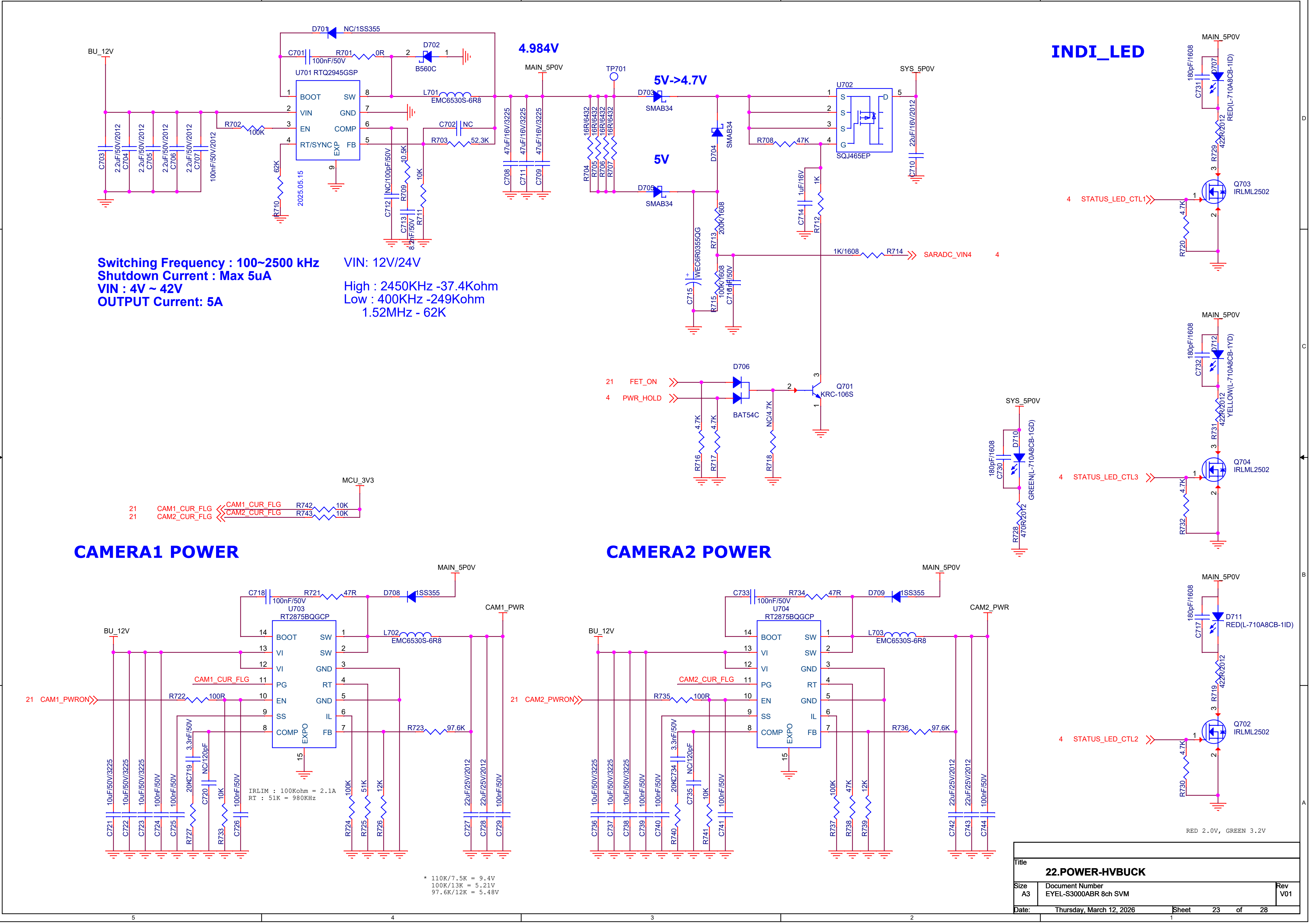

Title	22.POWER-HVBUCK
Size	A3
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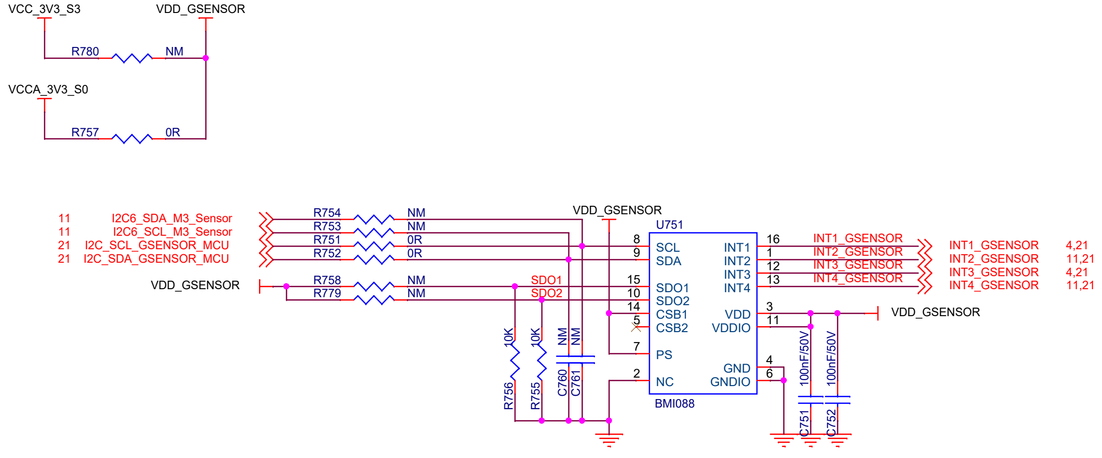
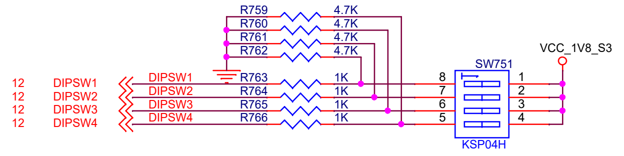
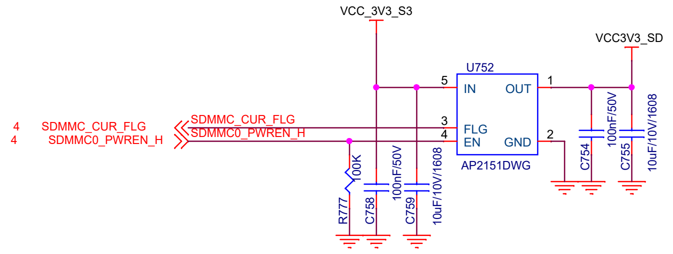
Digital, Triaxial Acceleration Sensor

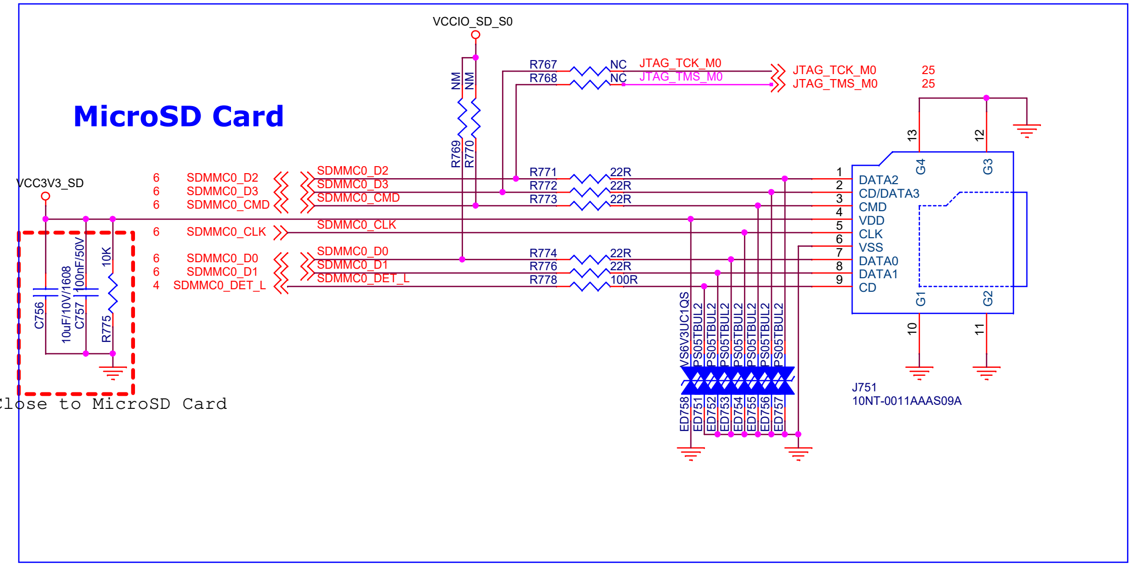



```
PS : 0= SPI, 1=I2C
I2C Addr
SDO1 : Acc 0= 0x18, 1=0x19
SDO2 : Gyro 0=0x68, 1=0x69
```





MicroSD Card



Title	23.MicroSD/GSENSOR
Size	A3
Document Number	EYEL-S3000ABR 8ch SVM
Rev	V01
Date:	Thursday, March 12, 2026
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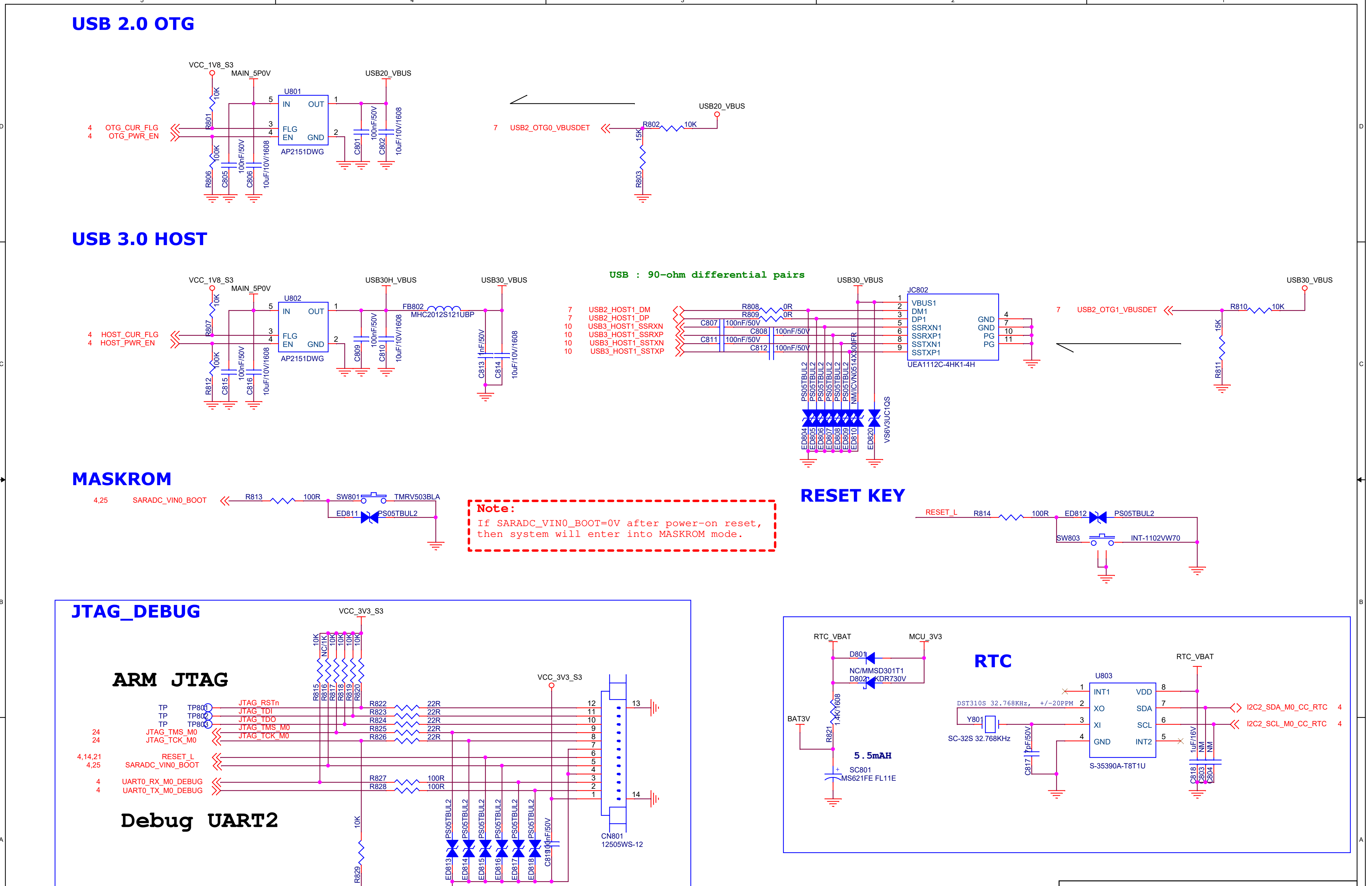
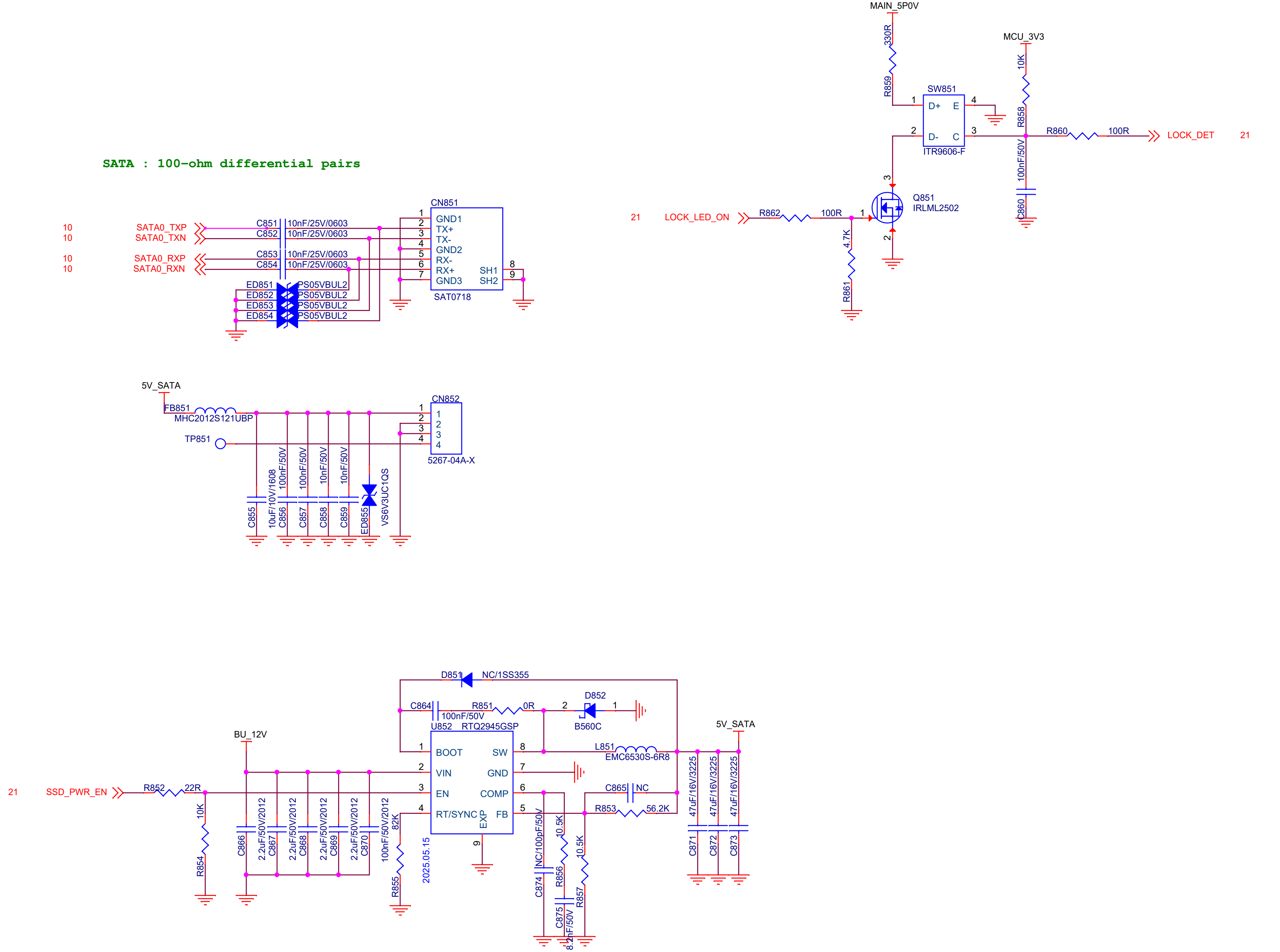


Title	24.DEBUG/USB2.0/USB3.0/RTC	
Size A3	Document Number EYEL-S3000ABR 8ch SVM	Rev V01
Date: Thursday, March 12, 2026	Sheet 25 of 28	

SATA



Switching Frequency : 100~2500 kHz
Shutdown Current : Max 5uA
VIN : 4V ~ 42V
OUTPUT Current: 5A

VIN: 12V/24V
RT/SYNC : 100K = 1MHz

Title	25.SATA
Size	A3
Document Number	EYEL-S3000ABR 8ch SVM
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Date:	Thursday, March 12, 2026
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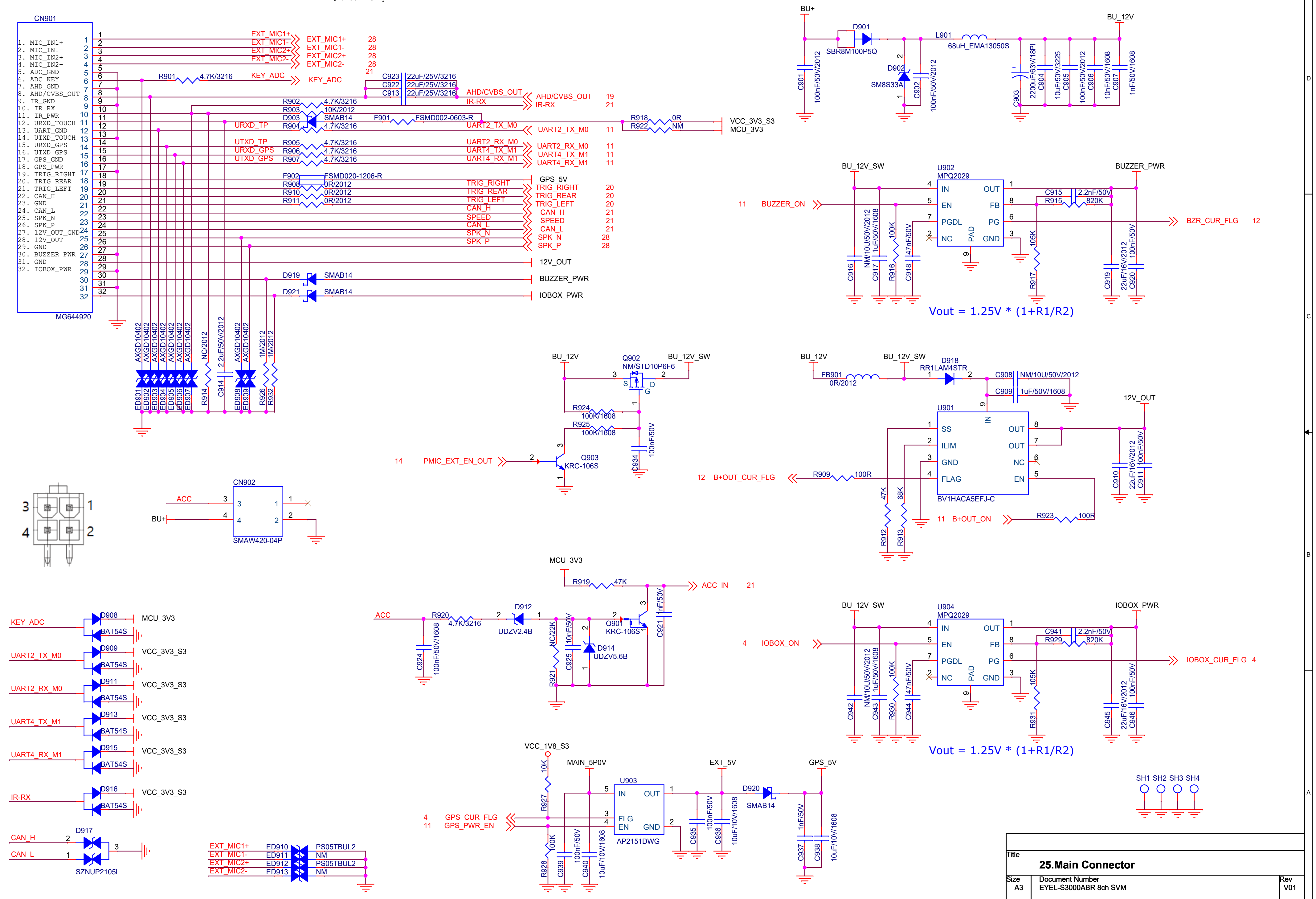
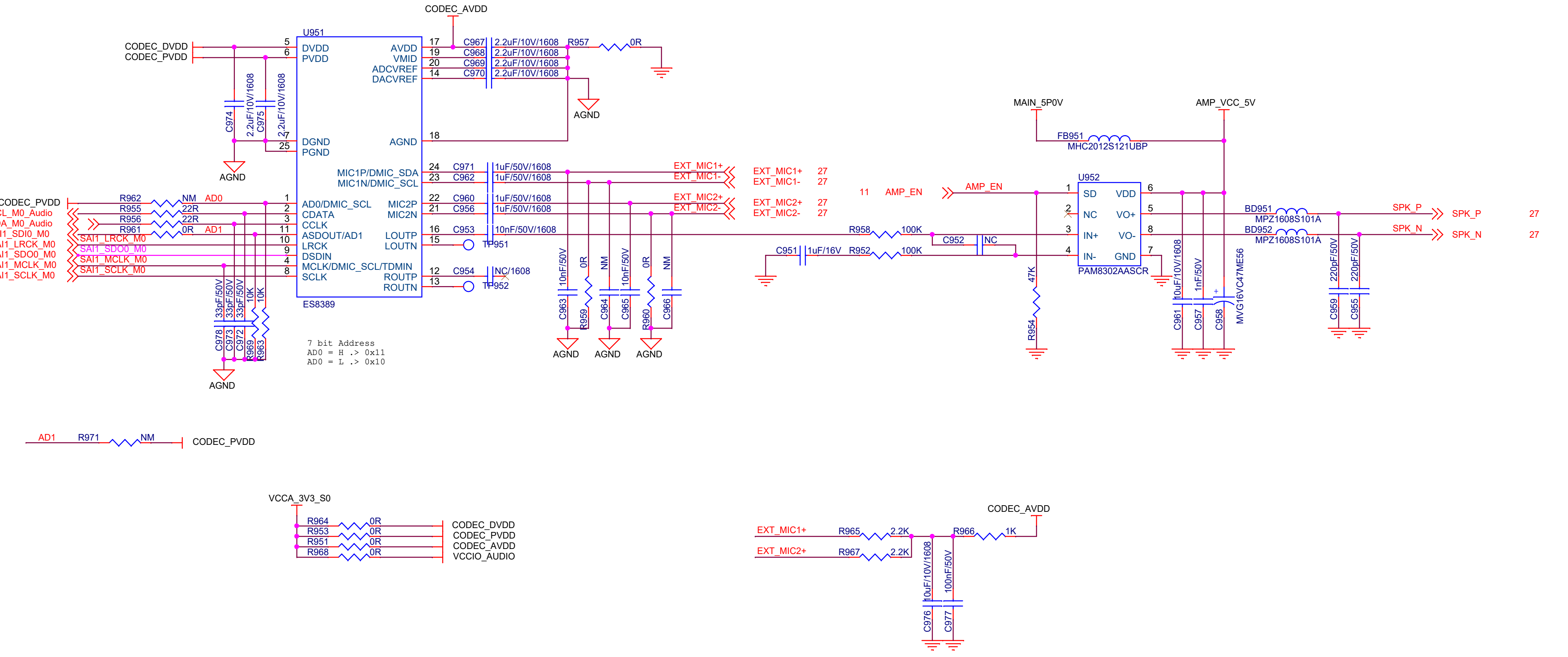
Main Connector

FSMD002-0603-R
- Ihold : 20mA
- Itrip : 60mA
@70 50% Itrip

UART2 : Ext Touch
UART4 : GPS



Title	25.Main Connector
Size	A3
Document Number	EYEL-S3000ABR 8ch SVM
Rev	V01
Date:	Thursday, March 12, 2026
Sheet	27 of 28



Title
25.Main Connector

Size	Document Number	Rev
A3	EYEL-S3000ABR 8ch SVM	V01

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